FLASHSIGN

DATABASE SCHEMA ↔️ FUNCTIONAL REQUIREMENTS TRACEABILITY & GAP ANALYSIS

Based on FlashSign SRS v4.0 Final Round 3 and the supplied FLASHSIGN DATABASE SCHEMA

Document control	Value
Analysis date	06 November 2026
Requirements baseline	FlashSign SRS v4.0 Final Round 3 (05 November 2026)
Schema source	User-uploaded schema.txt (1,267 lines; 100 table definitions detected)
Coverage	97 functional requirement IDs in the SRS functional-requirement traceability index
Purpose	Design review and gap discovery; not a migration script or final physical database design

Important interpretation: this report checks whether the proposed relational schema can persist and connect the information required by each SRS function. It does not certify that the backend APIs, permission checks, object storage, email provider, cryptographic signing, scheduled jobs or user interface are implemented.

1. Executive assessment

The proposed schema is broad and thoughtfully structured. It contains 100 table definitions spanning platform administration, multi-tenant organisations, users and roles, project management, tasks and planning, team collaboration, documents/contracts, electronic signatures, AI agents and SRS analysis.

At the domain level, every major functional module in the Round 3 SRS has corresponding persistence structures. The strongest areas are task/work tracking, signature workflow records, project hierarchy, role-based access configuration, and SRS-analysis candidate review.

However, table presence is not the same as complete requirement coverage. Several functions still depend on application logic, external services, cryptographic evidence, or additional persistence that is not explicit in the schema. Some high-impact ambiguities must be resolved before migrations are written.

Area	Assessment	Main observation
Authentication & onboarding	Mostly covered	Invitation, OTP, session, membership and role tables exist; clarify invitation/account transaction and verification semantics.
Role/access control	Strong structure; implementation-critical	Role settings and project membership exist; enforce owner override, tenant-safe FKs and server-side authorization.
Projects, tasks & planning	Strong	Core hierarchy, boards, tasks, work sessions, sprints and baseline history are represented.
Documents & contracts	Mostly covered	Document versions, shares, approvals and contracts exist; review file lifecycle, comments, and contract-signature state consistency.
Electronic signatures	Broad coverage; assurance gaps	Envelope, recipients, fields, tokens, evidence and event history exist; identity proofing, OTP and cryptographic certificate detail need decisions.
Teams & reporting	Mostly covered	Teams, meetings, time off, announcements and activity logs exist; several reports are derived and need precise scope rules.
AI/SRS analysis	Strong structure; external workflow gaps	Analysis runs, candidates, agent tokens, settings and events exist; provider secrets, prompt/model traceability and AI data-retention rules need review.

2. How to read the requirement mappings

- Required tables are the tables the function reads or writes in the normal workflow; optional tables support related UI conveniences, history, notifications or audit.
- Columns are listed as table.column. The table definitions themselves remain the source of truth for data types, allowed values and constraints.
- Relationships are expressed using the schema's foreign-key intent. The schema's multi-tenant rule requires tenant-to-tenant FKs to include organization_id, even when the abbreviated column listing shows the logical target.
- Status labels: Covered = core persistence is present; Partial = core tables exist but a requirement detail, constraint or supporting record is not explicit; Gap = no clear persistence structure is present; Derived = no extra table is required because the value should be calculated from source records.
- The SRS includes prototype evidence/status. A prototype marked implemented does not mean the production database or backend requirement is complete.

3. Cross-cutting database rules to preserve

- Every tenant-owned table must carry organization_id and every tenant query must be scoped to the active organisation. The schema states PostgreSQL RLS is required on every tenant table.
- Every FK from one tenant table to another must include organization_id as part of the reference; otherwise a row could point across tenants despite RLS.
- All tenant indexes should begin with organization_id. Uniqueness rules should be organisation-scoped unless the schema explicitly marks a global unique key.
- Files must remain in object storage. stored_file stores metadata and storage_key; never place PDF/image bytes or Base64 data in ordinary columns.
- Passwords, session tokens, invitations, signing tokens, verification codes and agent tokens must be stored only as hashes.
- Use transactions for invitation acceptance, task moves, signature state transitions, and candidate-to-task ingestion so related rows cannot become inconsistent.
- Append-only audit/event tables must be protected from update/delete by database permissions, not merely by UI convention.
- Derived counts, progress, health and expiry states should be calculated from source rows unless the SRS explicitly requires a historical snapshot.

4. Requirement-by-requirement mapping

The following register covers all 97 FR identifiers listed in the SRS. The SRS requirement title and description are retained as source-derived summaries; the table mapping and gap notes are database-design analysis.

Authentication, onboarding & access

FR-AUTH-001 — User Registration Form

SRS behaviour: The system shall create a user account only from a valid invitation: an organisation invitation sent by a Super Admin to the organisation's Owner (FR-AUTH-009), or a member invitation sent by the Owner or a Team Management role (FR-AUTH-004). The login page offers no open sign-up.

Actor: Invited Owner or Invited Member

Precondition: The visitor holds a pending, unexpired invitation link.

Trigger: Opening the invitation link (`/invite/[token]`).

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
organization_invitation	organization_invitation_id (PK), organization_id, invited_by_user_id, accepted_by_user_id, organization_name, owner_name, owner_email, plan, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	

invitation_project	invitation_project_id (PK), organization_id, invitation_id, project_id	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
verification_code	verification_code_id (PK), user_id, purpose, code_hash, attempts, expires_at, consumed_at, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: schema supports both invitation types, but the account-creation flow must atomically create/link users, org_member, role assignment and invitation status. Confirm that organisation_invitation acceptance creates the organisation when required.

SRS acceptance criteria: No account or organisation can be created without a valid invitation; accepted invitations join the correct organisation with the invited role.

FR-AUTH-002 — Email OTP Verification

SRS behaviour: The system shall verify the ownership of the user's email address using a 6-digit bcrypt-hashed OTP valid for 10 minutes.

Actor: Unverified User

Precondition: An unverified user record exists with an active OTP dispatch.

Trigger: User submits the 6-digit code on `/verify-email`.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
verification_code	verification_code_id (PK), user_id, purpose, code_hash, attempts, expires_at, consumed_at, created_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id,	

	detail, ip_address, user_agent, created_at	

Assessment: Partial: verification_code supports hashed OTP, expiry, attempts and consumption. The SRS says a successful invitation may itself verify email or use OTP; define which path is authoritative and ensure resend cooldown is enforced transactionally.

SRS acceptance criteria: Incorrect codes are rejected, maximum 5 attempts are enforced, and successful verification authenticates the session.

FR-AUTH-003 — 4-Step Org Onboarding

SRS behaviour: The system shall guide an Organization Owner, after accepting the Super Admin's organisation invitation, through a 4-step onboarding wizard to complete their organisation workspace.

Actor: Registered User (Organization Owner)

Precondition: User is authenticated as the Owner of an organisation that a Super Admin created or invited (FR-AUTH-009), and `organizationSetupCompleted` is false.

Trigger: Landing on `/onboarding` post-registration.

Table	Relevant columns currently defined	Role in this requirement
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
role_template	role_template_id (PK), role_key, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, sort_order	
department	department_id (PK), organization_id, name, sort_order, created_at	
user_preference	user_preference_id (PK), user_id, theme, default_calendar_view, default_planning_view, in_app_notifications, email_notifications, desktop_notifications, weekly_digest, updated_at	
organization_feature	organization_feature_id (PK), organization_id, feature_key, is_enabled, updated_by_user_id, updated_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: organisation profile/preferences and initial Owner membership are represented. The four-step setup-completion state is not explicit; decide whether setup is derived or add organization.setup_completed_at / onboarding_state. Personal preferences are global in user_preference; organisation timezone/working hours belong on organization.

SRS acceptance criteria: Wizard state persists across step navigation and accurately populates `currentOrganization` upon completion.

FR-AUTH-004 — Tokenized Team Invites

SRS behaviour: The system shall allow the Organization Owner and Team Management roles to invite members by email with one of the organisation's roles and, for roles with Project User Access, the projects assigned to them, allowing invitees to join via a secure token link without creating an organization.

Actor: Organization Owner or Team Management role (Sender); Invited User (Recipient)

Precondition: Inviting user's role has Team Management access. Team Member roles see no invite action, and `/teams/members/new` shows Access Denied.

Trigger: Submission of the \"Invite Member\" modal in `/teams/members`.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
organization_invitation	organization_invitation_id (PK), organization_id, invited_by_user_id, accepted_by_user_id, organization_name, owner_name, owner_email, plan, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
invitation_project	invitation_project_id (PK), organization_id, invitation_id, project_id	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
verification_code	verification_code_id (PK), user_id, purpose, code_hash, attempts, expires_at, consumed_at, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at,	

	bounced_at, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: invitation and invitation_project support role and project assignment. Add/confirm invitee email normalisation, role/project tenant consistency, and atomic acceptance. Consider an explicit expired state only if the UI must persist it; current schema derives expiry from expires_at.

SRS acceptance criteria: Accepted invitations join the correct organization and revoke further use of the token.

FR-AUTH-005 — Login & JWT Session

SRS behaviour: The system shall authenticate existing users using email and password, issuing a secure session cookie.

Actor: Registered User

Precondition: User has a registered, verified account.

Trigger: Submission of credentials at `/login`.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	

Assessment: Partial: session has token_hash, expiry, revocation and active organisation. SRS calls out JWT/session-cookie behaviour; clarify whether the database stores opaque session tokens or JWT refresh/session identifiers. Add session rotation/reuse detection only if required by chosen auth design.

SRS acceptance criteria: Valid credentials establish session; invalid credentials display generic error preventing user enumeration.

FR-AUTH-006 — Password Reset via OTP

SRS behaviour: The system shall allow users to reset forgotten passwords by verifying a 6-digit reset code delivered via email.

Actor: Registered User

Precondition: User account exists.

Trigger: User submits password reset request at `/forgot-password`.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
verification_code	verification_code_id (PK), user_id, purpose, code_hash, attempts, expires_at, consumed_at, created_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	

Assessment: Partial: verification_code purpose=password_reset supports the flow. Confirm invalidation of older codes, password-reset audit events, rate limits by account/IP and whether reset revokes all existing sessions.

SRS acceptance criteria: Password reset terminates existing device sessions and enforces minimum 8-character password security.

FR-AUTH-007 — Permission-Based Navigation & Route Protection

SRS behaviour: The system shall decide navigation items, pages, routes and actions (create, edit, delete, view) from the signed-in user's permissions: Super Admin or organisation member, Owner, Dashboard Access, Project Access and assigned projects, My Tasks access and Team Access (Section 2.5), using one central set of permission checks.

Actor: Any authenticated user

Precondition: User holds an active session.

Trigger: Rendering the sidebar, opening any page, or showing any action.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system,	

	admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: roles and memberships persist permissions. The schema does not itself enforce route/API checks; RLS alone is not sufficient for role permissions. Every endpoint/action must run central authorization logic and owner override.

SRS acceptance criteria: For each demo account, every restricted page listed in the Main Flow is hidden from navigation and shows Access Denied by direct URL; allowed pages open.

FR-AUTH-008 — Demo Accounts and Persona Switcher (Prototype)

SRS behaviour: For prototype demonstrations, the login page shall list six demo accounts (Super Admin, Owner, Project Manager, Team Lead, Developer, QA), each with predefined mock permissions and project assignments, and the header user menu shall let evaluators switch between them without logging out.

Actor: Demonstration Evaluator

Precondition: Prototype build.

Trigger: Selecting a demo account on `/login`, or choosing an account under Switch Demo Persona in the header user menu.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	

platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Prototype-only: demo accounts/persona switching should not be a production feature. If retained for test environments, isolate seeded identities and reset operations from production data.

SRS acceptance criteria: Each of the six accounts signs in, lands on its starting page and sees only its permitted navigation; switching accounts updates navigation immediately.

FR-AUTH-009 — Super Admin Dashboard & Organization Management

SRS behaviour: The system shall provide Super Admins with a separate dashboard and navigation to view, create, edit and delete organisations, invite organisations (their Owner) to FlashSign, manage those invitations, and set each organisation's plan and status.

Actor: Super Admin

Precondition: User is signed in as a Super Admin.

Trigger: Opening the Super Admin Dashboard (`/super-admin`).

Table	Relevant columns currently defined	Role in this requirement
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
organization_invitation	organization_invitation_id (PK), organization_id, invited_by_user_id, accepted_by_user_id, organization_name, owner_name, owner_email, plan, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Partial: platform_admin, organization, organization_invitation and platform_activity_log support Super Admin administration. Support-access approval is represented by support_access_grant; define whether every support session writes both platform and tenant audit records.

SRS acceptance criteria: A Super Admin can create, invite, edit and delete organisations and see their status; a Super Admin cannot open organisation pages; organisation users cannot open the Super Admin Dashboard.

FR-AUTH-010 — Dashboard Access & Switching

SRS behaviour: The system shall give each user the dashboards their role allows (Admin Dashboard at `/dashboard`, My Status at `/projects/home`), open the default one after login, and show a dashboard switcher in the header only when both are allowed.

Actor: Any organisation user

Precondition: User's role has Dashboard Access set (Section 2.5).

Trigger: Login, or selecting a dashboard in the header switcher or the sidebar.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: role flags support dashboard choice and project_member scopes project lists. DEC-28 remains open: decide whether Admin Dashboard metrics are organisation-wide or assigned-project-only; this affects queries and authorization, not necessarily new tables.

SRS acceptance criteria: Owner, Project Manager and Team Lead land on the Admin Dashboard and can switch to My Status; Developer and QA Tester land on My Status and cannot open `/dashboard`.

Projects & portfolio

FR-PROJ-001 — Project Creation

SRS behaviour: The system shall allow authorized users to create a new project workspace with a unique project key, name, owner, dates, tags, and priority.

Actor: Users whose role has Project Management Access (Organization Owner and Project Manager by default)

Precondition: User belongs to the active organization and their role has Project Management Access. Others do not see New Project, and `/projects/new` shows Access Denied ("Project Creation Restricted").

Trigger: Submission of the \"New Project\" modal at `/projects` or `/workspace/new`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
project_goal	project_goal_id (PK), organization_id, project_id, goal, sort_order	
project_tag	project_tag_id (PK), organization_id, project_id, tag	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: project plus project_member, board/board_column and optional project_goal/project_tag support creation. Confirm one default/main board and initial columns are created in the same transaction.

SRS acceptance criteria: New project appears in project portfolio listing and can receive tasks immediately.

FR-PROJ-002 — Project Overview Dashboard

SRS behaviour: The system shall render a comprehensive project dashboard displaying health status, sprint counts, task completion percentages, member avatars, urgent issues, and recent audit activity.

Actor: Authenticated user with access to the project (Project Management Access: any project; Project User Access: assigned projects only, FR-PROJ-010). Duplicate and Archive actions appear only for Project Management Access.

Precondition: Project ID exists.

Trigger: Navigation to `/projects/[projectId]`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	

project_goal	project_goal_id (PK), organization_id, project_id, goal, sort_order	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Derived: project overview metrics should be calculated from project/task/phase/epic/story/work_session/baseline rows. Avoid storing duplicated counts; define the metric formulas and access scope.

SRS acceptance criteria: Metric cards accurately reflect the sum of underlying task statuses and estimated work effort.

FR-PROJ-003 — Dynamic Project Health

SRS behaviour: The system shall calculate project health (`on_track`, `at_risk`, `delayed`) automatically from the project timeline (REQ-PRJ-006), while permitting a Project Manager to set `off_track` manually.

Actor: System (Automated); Project Manager (Manual Override)

Precondition: Project exists.

Trigger: Task schedule changes or manual setting in Project Settings.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id,	

	phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	

Assessment: Partial: health_mode and manual health are stored; auto health is derived from schedule/task/work-session data. The exact health algorithm and thresholds need to be specified and tested.

SRS acceptance criteria: Moving task due dates beyond project target automatically updates health badge from `on_track` to `at_risk` or `delayed` in auto mode.

FR-PROJ-004 — Custom Board Columns

SRS behaviour: The system shall allow teams to rename, reorder, add, and recolor Kanban columns while mapping every column to one of five fixed task stages (`todo`, `in_progress`, `review`, `done`, `blocked`).

Actor: Project Manager, Engineering Lead

Precondition: User possesses project management authority.

Trigger: Editing columns in Project Settings $\rightarrow$ Board.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section,	

	is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: board_column has status, color and sort_order. Enforce required workflow stages for non-mirroring boards and prevent deleting/moving columns that would orphan tasks.

SRS acceptance criteria: Custom column names appear on the Kanban board; cards dropped in custom columns adopt the underlying stage status.

FR-PROJ-005 — Multi-Board Support

SRS behaviour: The system shall allow a project to maintain secondary boards (e.g., \"Testing Board\", \"Deployment Board\") where cards move between boards while retaining unified history and comments.

Actor: Project Manager, Engineering Lead

Precondition: Project exists.

Trigger: Adding an extra board in Project Settings.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: board and board_column support multiple boards and mirrored boards. Clarify how task.board_id and board_column_id remain consistent when a board mirrors the main board.

SRS acceptance criteria: Moving a task to a secondary board creates a board transition record and isolates board-level stage durations.

FR-PROJ-006 — Project Archival & Soft Delete

SRS behaviour: The system shall permit project managers to archive completed or inactive projects, hiding them from active portfolio views while preserving complete audit history.

Actor: Users whose role has Project Management Access

Precondition: Project exists.

Trigger: User selects Archive on the project overview, or Archive or Delete in the Danger Zone of project Settings (`/projects/[projectId]/settings`).

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: project.deleted_at and archived_at support archive/soft delete. The schema has no project-specific trash/restore audit entity; activity_log can record this. Define cascade/retention behaviour for linked tasks, files, documents and signed agreements.

SRS acceptance criteria: Archived projects do not appear in active task pickers, search dropdowns, or active workload aggregations.

FR-PROJ-007 — Phase $\rightarrow$ Epic $\rightarrow$ Story Plan

SRS behaviour: The system shall let each project choose which plan levels it uses: Phases (stored as Milestones), Epics, Stories, Tasks, Subtasks and Backlog. Presets are Full (Phases › Epics › Stories › Tasks › Subtasks, default), Agile (Epics › Stories › Tasks › Subtasks), Phased (Phases › Epics › Tasks › Subtasks), Simple phases (Phases › Tasks › Subtasks) and Tasks only (Tasks › Subtasks). Each level is created inside the nearest level above it that the project uses, and tasks sit in the lowest level used. Levels are chosen in the New Project wizard and can be changed in Project Settings (see Annex B.2 §3.2).

Actor: Product Owner, Project Manager

Precondition: Project exists.

Trigger: Navigating to `/projects/[projectId]/plan` or assigning parent entities in task drawer.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project_goal	project_goal_id (PK), organization_id, project_id, goal, sort_order	

Assessment: Covered: phase, epic, story and task represent the hierarchy. Validate mutually exclusive lowest-plan-level links and same-project relationships.

SRS acceptance criteria: Effort and completion progress aggregate up from tasks to stories, epics, and milestones.

FR-PROJ-008 — Recent Projects History

SRS behaviour: The system shall maintain an ordered list of recently visited projects per user to facilitate rapid workspace switching.

Actor: Authenticated User

Precondition: User visits project routes.

Trigger: Landing on `/projects/[projectId]`.

Table	Relevant columns currently defined	Role in this requirement
user_recent_project	user_recent_project_id (PK), organization_id, user_id, project_id, last_opened_at	
user_favorite_project	user_favorite_project_id (PK), organization_id, user_id, project_id, created_at	

project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	

Assessment: Covered: user_recent_project and user_favorite_project persist per-user project navigation. Confirm cleanup when a project is permanently deleted.

SRS acceptance criteria: Visiting a project immediately places it at the top of recent project shortcuts.

FR-PROJ-009 — Portfolio Search & Filter

SRS behaviour: The system shall provide real-time debounced search, status filtering, category grouping, and sorting across the projects the user may see: every organisation project for Project Management Access ("All Projects"), and assigned projects only for Project User Access ("My Assigned Projects").

Actor: Authenticated User

Precondition: Navigation to `/projects`.

Trigger: Entering search keywords or selecting filter chips.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
project_tag	project_tag_id (PK), organization_id, project_id, tag	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	

Assessment: Covered: project fields, project_tag and membership allow search/filter. Add indexes for commonly filtered fields (organization_id, status, priority, project_key, name) as query design requires.

SRS acceptance criteria: Search query matches both project keys and titles with debounced responsive rendering.

FR-PROJ-010 — Project Access Scope

SRS behaviour: The system shall limit what a user can see and do in projects according to their role's Project Access and, for User Access, their assigned projects.

Actor: Any organisation user

Precondition: User's role has Project Access set (Section 2.5); User Access users have assigned projects (FR-AUTH-004, FR-TEAM-008).

Trigger: Opening the project list or any project page.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id,	

	name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
invitation_project	invitation_project_id (PK), organization_id, invitation_id, project_id	

Assessment: Partial: project_access role plus project_member rows implement management vs assigned-project scope. Ensure project membership is automatically removed/updated when a member is suspended, removed or role changes.

SRS acceptance criteria: Team Lead, Developer and QA Tester see only their assigned projects, can work inside them, see Settings read-only, and are blocked from Reports, the Danger Zone, project creation and unassigned projects; Project Manager can create, edit and delete any project.

Tasks & Kanban

FR-TASK-001 — Task Key Generation

SRS behaviour: The system shall create tasks with auto-incremented project-prefixed keys (e.g., \"FS-101\"), assigning priority, type, estimated duration, unit, dates, and reporter.

Actor: Any authenticated team member

Precondition: Active project exists.

Trigger: Submission of `CreateTaskModal` or Quick Create button.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	

Assessment: Covered: project.next_task_number and task.task_key support non-reused keys. Allocate numbers atomically under concurrent creation; unique(project_id, task_key) is required.

SRS acceptance criteria: Task keys are sequential, unique per project, and persist across board views.

FR-TASK-002 — Kanban Drag-and-Drop

SRS behaviour: The system shall allow users to transition tasks between board columns using accessible drag-and-drop powered by `@dnd-kit`, instantly updating status and triggering work session timers.

Actor: Developer, QA Tester, Project Manager

Precondition: Tasks exist in project.

Trigger: Dragging a card and dropping it into a target column on `/projects/[projectId]/board`.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: task.board_id, board_column_id, status and rank support drag/drop. A move must update status/column/rank and append task_status_change in one transaction.

SRS acceptance criteria: Dragging a card across columns updates status in state, records audit timestamp, and starts/stops timers accurately.

FR-TASK-003 — Subtask Schedule Derivation

SRS behaviour: The system shall support creating subtasks linked via `parentId`, calculating parent duration and dates strictly from the subtasks to avoid double-counting work effort.

Actor: Developer, Project Manager

Precondition: Parent task exists.

Trigger: Adding a subtask from the Task Detail Drawer or tasks list.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	

Assessment: Partial: parent_task_id and date/duration fields support subtask scheduling. The SRS derivation rules need a precise policy for parent date recalculation and manual overrides.

SRS acceptance criteria: Adding subtasks to a 5-day parent task causes project total workdays to reflect subtask durations rather than adding parent plus subtasks.

FR-TASK-004 — Container Timer Suppression

SRS behaviour: The system shall treat a task with subtasks as a organizational container, disallowing active timers on the parent task and requiring timers to run on individual subtasks.

Actor: System / Developer

Precondition: Task possesses one or more subtasks (`hasSubtasks = true`).

Trigger: Task is transitioned into `in_progress`.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
parent_task_id	NOT FOUND IN SCHEMA	Gap: table name is referenced by mapping but absent
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	

Assessment: Partial: task hierarchy and work_session support timer suppression for container tasks. This is application logic; add a database constraint or service validation preventing work sessions against parent/container tasks if required.

SRS acceptance criteria: Moving a container task to In Progress starts 0 timers; moving its subtask to In Progress starts an active timer.

FR-TASK-005 — Guarded Move Dialog

SRS behaviour: When a user attempts to transition a parent task to `Done` while one or more of its subtasks remain open, the system shall intercept the action and display a confirmation barrier.

Actor: Developer, QA Tester, Project Manager

Precondition: Parent task has open subtasks (`status !== 'done'`).

Trigger: Dragging or setting parent task to `done`.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: task_status_change captures move history, but the guarded move dialog's validation reasons/blocked transitions are not stored explicitly. Decide whether blocked-move attempts need an audit event.

SRS acceptance criteria: Attempting to complete a task with open subtasks presents dialog; cancelling leaves card in original status.

FR-TASK-006 — Task Detail Drawer Tabs

SRS behaviour: The system shall provide a slide-over drawer displaying tabs for Details, Checklist, Progress, Comments, Attachments, and Journey, allowing modification of all task attributes.

Actor: Authenticated User

Precondition: Task exists.

Trigger: Clicking a task card or row.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_comment	task_comment_id (PK), organization_id, task_id, author_user_id, body, via, created_at, updated_at	
task_checklist_item	task_checklist_item_id (PK), organization_id, task_id, text, is_completed, sort_order	
task_attachment	task_attachment_id (PK), organization_id, task_id, stored_file_id, created_at	
task_link	task_link_id (PK), organization_id, task_id, linked_task_id, link_type, created_by_user_id, created_at	
task_tag	task_tag_id (PK), organization_id, task_id, tag	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: task plus task_comment, checklist, attachment, link, tag, status history and work_session support detail tabs.

SRS acceptance criteria: Modifying attributes in drawer updates board card display and triggers activity audit logs.

FR-TASK-007 — Schedule Overrun Warning

SRS behaviour: The system shall dynamically compute whether an In Progress task has exceeded its estimated duration, displaying visible overrun badges and elapsed time warnings.

Actor: System

Precondition: Task is in `in_progress` status and possesses an `estimatedDuration`.

Trigger: Elapsed active work session time exceeds estimate.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics,	

	uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	

Assessment: Derived: schedule overrun warning can be calculated from due_date/start_date/estimated_duration and status. Persist a notification only when the SRS requires a delivered reminder; do not store a duplicate warning flag.

SRS acceptance criteria: Tasks whose active time exceeds estimate display overrun indicators and appear in overrun-only filter views.

FR-TASK-008 — Multi-Board Task Journey

SRS behaviour: The system shall record each movement of a task across project boards, detailing entry time, exit time, actor, source (web or AI agent), comments posted during the stage, and time logged on each board.

Actor: Developer, QA Tester, Project Manager

Precondition: Project has more than one board configured (`boards.length > 1`).

Trigger: Task card moved to another board.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: task board journey and status history are represented by task_status_change, including from/to board and column.

SRS acceptance criteria: A card moved from Development Board to Testing Board displays distinct time logged in Dev vs Testing.

FR-TASK-009 — Task Checklists

SRS behaviour: The system shall allow users to attach structured checklist items to tasks, toggling completion status and rendering progress indicators on board cards.

Actor: Authenticated User

Precondition: Task drawer is open.

Trigger: Adding or checking checklist items in Checklist tab.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_checklist_item	task_checklist_item_id (PK), organization_id, task_id, text, is_completed, sort_order	

Assessment: Covered: task_checklist_item stores each checklist item and completion state. Enforce stable sort_order and cascade rules.

SRS acceptance criteria: Checking all checklist items marks checklist progress at 100% on card.

FR-TASK-010 — Timestamped Task Comments

SRS behaviour: The system shall allow team members to post timestamped comments on tasks, attributing authorship to the authenticated user.

Actor: Authenticated User

Precondition: Task drawer is open.

Trigger: Submitting comment in Comments tab.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_comment	task_comment_id (PK), organization_id, task_id, author_user_id, body, via, created_at, updated_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	

Assessment: Covered: task_comment records author, body and timestamps. If edit/delete history is required, current row updated_at is insufficient; add comment revision/audit history.

SRS acceptance criteria: Posted comments display author name, formatted relative time, and persist across drawer sessions.

FR-TASK-011 — Task Multi-Filter & Search

SRS behaviour: The system shall provide search and filtering across tasks by keyword, status, priority, milestone, epic, assignee, and overrun state.

Actor: Authenticated User

Precondition: Navigation to `/projects/[projectId]/tasks`.

Trigger: User changes filter dropdowns or enters search text.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
task_tag	task_tag_id (PK), organization_id, task_id, tag	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	

Assessment: Covered: task fields, task_tag, project and assignee/reporter support filters. Add search indexes/full-text search if volume requires it.

SRS acceptance criteria: Filtering by an assignee shows only tasks assigned to that individual while preserving parent-subtask relationships.

FR-TASK-012 — Personal \"My Tasks\" Queue

SRS behaviour: The system shall aggregate all tasks assigned to the authenticated user across all organization projects into a centralized personal workspace view.

Actor: Users whose role has My Tasks access (all default roles)

Precondition: User has assigned tasks across one or more projects.

Trigger: Navigation to `/my-tasks` or `/projects/my-tasks`.

Table	Relevant columns currently defined	Role in this requirement
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status,	

	health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	

Assessment: Partial: My Tasks can query assigned tasks and work_session using org_member/role flags. If the queue includes followed/mentioned tasks, add task_watchers or task_mention records; the current schema has no watcher/mention relation.

SRS acceptance criteria: Personal view aggregates tasks across multiple distinct projects accurately reflecting live timer state.

Planning & timeline

FR-PLAN-001 — Forward Schedule Engine

SRS behaviour: The system shall dynamically compute project completion dates, total workdays, and schedule variances by evaluating task estimates, subtask spans, and dependencies without double-counting effort.

Actor: System

Precondition: Project tasks possess estimated durations and start dates.

Trigger: Creation, update, or duration change of any task or milestone.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key,	

	title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	

Assessment: Covered: project/phase/epic/story/task date and duration fields support forward scheduling. Define dependency recalculation rules and manual date override behaviour.

SRS acceptance criteria: Increasing a task estimate expands calculated project end date and increments calendar variance days.

FR-PLAN-002 — Baseline History Versioning

SRS behaviour: The system shall allow project managers to re-baseline the project schedule, archiving the previous start/end dates with a mandatory justification reason in a historical audit log.

Actor: Project Manager, CEO

Precondition: Project possesses an existing baseline.

Trigger: Clicking \"Re-baseline Project Schedule\" on `/projects/[projectId]/timeline`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: baseline_change records old/new project dates and reason. If baseline changes can be made at phase/epic/task level, the current table is too narrow; extend it with resource_type/resource_id or separate baseline detail rows.

SRS acceptance criteria: Baseline history modal displays chronological log of all re-baselining events with user attribution and reason.

FR-PLAN-003 — Sprint Planning & Lifecycle

SRS behaviour: The system shall support sprint creation, goal definition, date range allocation, capacity setting, and lifecycle transitions (`planned`, `active`, `completed`).

Actor: Product Owner, Project Manager

Precondition: Project exists.

Trigger: Creating or starting a sprint on `/projects/[projectId]/sprints`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics,	

	uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
sprint	sprint_id (PK), organization_id, project_id, name, goal, status, start_date, end_date, capacity, capacity_unit, started_at, completed_at, created_at, updated_at	
sprint_task	sprint_task_id (PK), organization_id, sprint_id, task_id, added_at, removed_at, outcome	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	

Assessment: Covered: sprint, sprint_task and task.sprint_id support sprint lifecycle and scope history. Ensure sprint_task records each add/remove event and outcomes without overwriting prior history.

SRS acceptance criteria: Incomplete tasks from completed sprints automatically roll over to backlog or designated next sprint.

FR-PLAN-004 — Unified Planning Calendar

SRS behaviour: The system shall provide an interactive calendar aggregating task deadlines, milestone due dates, team meetings, time-off periods, and planning reminders.

Actor: Authenticated User

Precondition: Calendar events or dated tasks exist.

Trigger: Navigation to `/planner/calendar`.

Table	Relevant columns currently defined	Role in this requirement
calendar_event	calendar_event_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, category, start_at, end_at, all_day, location, meeting_link, created_at, updated_at	
calendar_event_attendee	calendar_event_attendee_id (PK), organization_id, calendar_event_id, user_id	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
meeting	meeting_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, start_at, end_at, location,	

	meeting_link, status, created_at, updated_at	
meeting_participant	meeting_participant_id (PK), organization_id, meeting_id, user_id	
time_off_request	time_off_request_id (PK), organization_id, user_id, decided_by_user_id, leave_type, start_date, end_date, duration_days, is_partial_day, reason, notes, status, decision_note, decided_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	

Assessment: Partial: unified calendar can read tasks, phases, meetings and time off. calendar_event stores custom events; meeting is a separate entity. Clarify whether all-day events, recurrence, reminders and external calendar sync are required; recurrence is not modelled.

SRS acceptance criteria: Dates updated on tasks immediately reflect on the planning calendar without manual refresh.

FR-PLAN-005 — Milestone Progress Rollup

SRS behaviour: The system shall support defining project milestones (phases) with target dates, computing milestone progress automatically based on linked task completion.

Actor: Project Manager, Product Owner

Precondition: Project exists.

Trigger: Navigation to `/projects/[projectId]/milestones`.

Table	Relevant columns currently defined	Role in this requirement
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	

Assessment: Derived: milestone/phase progress is calculated from linked epics/stories/tasks. Define weighting and completion rules; phase status itself is intentionally not stored.

SRS acceptance criteria: Completing tasks linked to a milestone advances the milestone progress percentage proportionally.

FR-PLAN-006 — Duration Unit Normalization

SRS behaviour: The system shall allow users to input estimates in days, weeks, or months while normalizing all internal schedule calculations into standard working days (1 week = 5 workdays, 1 month = 20 workdays).

Actor: Authenticated User

Precondition: Task or project estimate input.

Trigger: User toggles unit select in `EstimateField` or `DurationField`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	

Assessment: Partial: duration_unit fields exist on project/task/epic/story. Standardise units and conversion/rounding rules; phase has no duration_unit or estimated duration despite schedule features.

SRS acceptance criteria: Entering \"2 weeks\" stores 10 workdays in calculator and displays \"2 weeks\" in task details.

FR-PLAN-007 — Timeline Variance Chart

SRS behaviour: The system shall render a comparative Gantt chart illustrating the original baseline bar versus the current calculated schedule bar, highlighting drift.

Actor: Project Manager, Executive

Precondition: Project timeline calculated.

Trigger: Viewing `/projects/[projectId]/timeline`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key,	

	title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	

Assessment: Partial: baseline_change and date fields support variance charts. If chart must compare multiple baseline snapshots, current baseline_change stores only before/after values per change, not a named complete baseline snapshot.

SRS acceptance criteria: Chart accurately displays the visual difference between original baseline dates and calculated forecast dates.

SRS analysis

FR-SRS-001 — Project File Categorization

SRS behaviour: The system shall allow users to upload specification and project documents (`.pdf`, `.doc`, `.docx`, `.figma`, `.archive`, `.image`), tagging their category as `srs`, `test_report`, `doc`, or `other`.

Actor: Product Owner, Project Manager, Developer

Precondition: Active project exists.

Trigger: Dropping or selecting files on `/projects/[projectId]/files`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: project_file categorises SRS and related files and points to stored_file. Ensure extraction status is separate from file malware scan status.

SRS acceptance criteria: Uploaded files tagged with category `srs` expose the AI analysis action trigger.

FR-SRS-002 — Multi-Stage SRS Analysis

SRS behaviour: The system shall execute a multi-stage analysis pipeline across uploaded SRS files, synthesizing suggested user stories, acceptance criteria, story points, and priority classifications.

Actor: Product Owner, Project Manager

Precondition: Project file categorized as `srs` exists with `analysisStatus !== 'tasks_created'`.

Trigger: Clicking \"Analyze with AI\" in `SrsActionModal`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date,	

	original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag	

Assessment: Covered: srs_analysis_run stores each run and srs_task_candidate stores proposed output and provenance. Define supported file types/OCR extraction and provider failure retry behaviour.

SRS acceptance criteria: Analysis modal shows animated 4-step progress and transitions candidate tasks to the review stage.

FR-SRS-003 — SRS Review & Audit Modal

SRS behaviour: The system shall present an interactive review modal displaying all AI-extracted candidate requirements, enabling filtering by Accepted, Edited, and Rejected states with live metric tallies.

Actor: Product Owner, Project Manager, Lead Developer

Precondition: SRS file has completed analysis.

Trigger: Opening `SrsGeneratedTasksModal`.

Table	Relevant columns currently defined	Role in this requirement
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at,	

	updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: candidate decision/reviewer timestamps support human review. No explicit review-session/modal record or per-field edit history; add only if audit needs to reconstruct every edit.

SRS acceptance criteria: Status tab switches accurately isolate accepted, edited, and rejected task subsets.

FR-SRS-004 — In-Review Task Refinement

SRS behaviour: The system shall allow reviewers to inline-edit candidate task titles, descriptions, acceptance criteria, story points, priority, and type before ingestion, recording an `isEdited = true` audit flag.

Actor: Product Owner, Engineering Lead

Precondition: Candidate task displayed in Review Modal.

Trigger: Clicking \"Edit\" on a candidate task card.

Table	Relevant columns currently defined	Role in this requirement
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: candidate stores original and edited title/description/criteria/priority/type/points. This is a strong before/after model; confirm all editable fields have original_* counterparts.

SRS acceptance criteria: Edited tasks display \"Edited\" badge and preserve both original AI text and human-modified text.

FR-SRS-005 — Candidate Task Rejection

SRS behaviour: The system shall permit reviewers to reject candidate tasks extracted from an SRS, capturing an optional rejection justification and excluding the item from project board ingestion.

Actor: Product Owner, Project Manager

Precondition: Candidate task displayed in Review Modal.

Trigger: Clicking \"Reject\" on a candidate task card.

Table	Relevant columns currently defined	Role in this requirement
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: decision=rejected and rejection_reason support rejection. Require rejection_reason where policy demands it.

SRS acceptance criteria: Rejected items do not generate board tasks upon batch confirmation.

FR-SRS-006 — Batch Task Ingestion

SRS behaviour: The system shall batch-convert all accepted and edited candidate tasks into permanent project tasks, creating bidirectional traceability links between the project task and the source SRS document section.

Actor: Product Owner, Project Manager

Precondition: At least one candidate task has `outcome = 'accepted'`.

Trigger: Clicking \"Generate Tasks (N)\" in Review Modal footer.

Table	Relevant columns currently defined	Role in this requirement
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date,	

	acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: candidates can link to created_task_id and candidate tags. Batch commit should be transactional and idempotent; consider a batch/group identifier if one run can be committed in several batches.

SRS acceptance criteria: Generated tasks appear on Kanban board with a link back to the specific source document and section.

AI coding agent hub

FR-AGENT-001 — Multi-Agent Configuration

SRS behaviour: The system shall maintain configuration profiles and connection guides for four primary AI coding agent clients: Anthropic Claude Code, OpenAI Codex, Google Gemini CLI, and Google Antigravity.

Actor: Developer, Engineering Lead

Precondition: Navigation to `/ai-agents/connect`.

Trigger: Selecting an AI agent product card.

Table	Relevant columns currently defined	Role in this requirement
agent_provider	agent_provider_id (PK), organization_id, name, provider, model, is_configured, is_enabled, color	
agent_provider_scope	agent_provider_scope_id (PK), organization_id, agent_provider_id, scope	
agent_setting	agent_setting_id (PK), organization_id, is_enabled, idle_timeout_minutes, day_cutoff, allow_claim_unassigned, allow_server_ai_on_blocked, allow_llm_processing, updated_at	
agent_setting_allowed_agent	agent_setting_allowed_agent_id (PK), organization_id, agent_setting_id, agent_client, is_allowed	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	

Assessment: Partial: agent_provider and scopes/config flags exist, but provider credentials/secrets are not represented (they should normally live in a secret manager, not plaintext DB). Define connection test status and last-checked/error metadata if UI needs it.

SRS acceptance criteria: Setup instructions reflect the exact JSON configuration required by each vendor's MCP client.

FR-AGENT-002 — Personal Access Token Gen

SRS behaviour: The system shall allow developers to generate Personal Access Tokens (`pmat_...`) with user-defined names and expiration periods (30, 90, or 365 days), revealing the full plaintext token only once at creation.

Actor: Authenticated Developer / Engineer

Precondition: User holds developer permissions.

Trigger: Submitting \"Create Token\" dialog in `/ai-agents/tokens`.

Table	Relevant columns currently defined	Role in this requirement
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: agent_access_token stores only token_hash and last_four; usage is tracked per agent client. Show raw token only once and never recover it.

SRS acceptance criteria: Full token is accessible only during the creation modal; subsequent table views display masked `pmat_••••••••1234`.

FR-AGENT-003 — Agent Token Revocation

SRS behaviour: The system shall permit users to revoke any active personal access token, immediately rejecting subsequent tool calls from connected AI agents.

Actor: Authenticated User, Org Admin

Precondition: Token exists in `active` state.

Trigger: Clicking \"Revoke\" in `/ai-agents/tokens`.

Table	Relevant columns currently defined	Role in this requirement
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent,	

	prev_hash, hash, created_at	

Assessment: Covered: revoked_at/revoked_by_user_id support revocation. Add explicit reason/audit metadata if required; ensure revocation is checked on every MCP request.

SRS acceptance criteria: Revoked token immediately halts further MCP actions in API middleware.

FR-AGENT-004 — Live Work Session Timer

SRS behaviour: The system shall automatically start a work session timer when a task is moved to `in_progress` (by human or AI agent), pause any preceding running timers for that user, and stop the timer when moved to `review` or `done`.

Actor: System / AI Agent / Developer

Precondition: Task is a leaf task (`hasSubtasks = false`).

Trigger: Task status transitions into or out of `in_progress`.

Table	Relevant columns currently defined	Role in this requirement
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
agent_setting	agent_setting_id (PK), organization_id, is_enabled, idle_timeout_minutes, day_cutoff, allow_claim_unassigned, allow_server_ai_on_blocked, allow_llm_processing, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: work_session supports running/ended sessions, source and task/project context. Enforce one running session per user and non-overlap as specified.

SRS acceptance criteria: Moving task A to In Progress starts timer A; moving task B to In Progress automatically stops timer A with `switched_task` and starts timer B.

FR-AGENT-005 — Idle Timeout & Auto-Stops

SRS behaviour: The system shall enforce automatic timer shutdowns when an active session exceeds the organization idle timeout or crosses the daily cutoff threshold (e.g., 20:00), flagging the session with `needsReview = true`.

Actor: System (Scheduler / State Evaluator)

Precondition: Open work session exists.

Trigger: Time elapses past `idleTimeoutMinutes` without agent activity, or current time exceeds `dayCutoff`.

Table	Relevant columns currently defined	Role in this requirement
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
agent_setting	agent_setting_id (PK), organization_id, is_enabled, idle_timeout_minutes, day_cutoff, allow_claim_unassigned, allow_server_ai_on_blocked, allow_llm_processing, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: agent_setting and work_session support idle timeout and day cutoff. Need persist the last activity heartbeat reliably and define job execution/retry/clock semantics.

SRS acceptance criteria: Abandoned timers shut down automatically and display `needsReview` banners on the timesheet view.

FR-AGENT-006 — MCP Tool Execution

SRS behaviour: The system shall expose a formalized tool contract over MCP allowing AI agents to list assigned tasks, start work sessions, record progress notes, continue sessions, and mark tasks for review.

Actor: AI Coding Agent Client

Precondition: Valid PAT provided in MCP session initialization.

Trigger: JSON-RPC tool invocation from agent.

Table	Relevant columns currently defined	Role in this requirement

agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task_comment	task_comment_id (PK), organization_id, task_id, author_user_id, body, via, created_at, updated_at	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at	

Assessment: Partial: agent_event logs tool actions and outcomes; domain tables persist task/time changes. If MCP JSON-RPC request/response replay or debugging is required, add request_id/correlation_id and protocol version fields.

SRS acceptance criteria: Agent tool calls execute exact state transitions identical to web UI button clicks.

FR-AGENT-007 — Agent Telemetry Feed

SRS behaviour: The system shall record and display an immutable activity feed of all actions performed by AI agents (`listed_tasks`, `started`, `continued`, `paused`, `note`, `finished`, `time_logged`, `auto_stopped`).

Actor: Developer, Engineering Lead, Auditor

Precondition: Agent tool executions occur.

Trigger: System receives agent tool call.

Table	Relevant columns currently defined	Role in this requirement
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at	
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: agent_event plus agent_token_usage provide telemetry. Define retention and redact sensitive payloads from detail.

SRS acceptance criteria: Agent activity feed accurately distinguishes between human web actions and agent CLI actions.

FR-AGENT-008 — AI Privacy Blocked Projects

SRS behaviour: The system shall allow organization administrators to establish global AI policies, including a blacklist of `blockedProjectIds` whose tasks and data may never be accessed by or transmitted to AI coding agents.

Actor: Organization Owner

Precondition: User is the Organization Owner. The Organisation tab of AI Agent Hub is hidden from all other roles and `/ai-agents/organisation` shows Access Denied ("Owner Authority Required"); other AI Agent Hub tabs stay available.

Trigger: Updating AI policies in `/ai-agents/organisation`.

Table	Relevant columns currently defined	Role in this requirement
agent_setting	agent_setting_id (PK), organization_id, is_enabled, idle_timeout_minutes, day_cutoff, allow_claim_unassigned, allow_server_ai_on_blocked, allow_llm_processing, updated_at	
agent_blocked_project	agent_blocked_project_id (PK), organization_id, agent_setting_id, project_id	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at	

Assessment: Partial: agent_blocked_project and agent_setting flags represent project restrictions and external LLM consent. Define precedence for blocked project vs server AI, and log every attempted/allowed access.

SRS acceptance criteria: Tasks belonging to blocked projects never appear in agent `list_tasks` output.

FR-AGENT-009 — Agent Timesheet Reporting

SRS behaviour: The system shall aggregate work sessions into a timesheet view distinguishing hours logged via web UI, manual entry, Claude Code, Codex, Gemini CLI, and Antigravity.

Actor: Developer, Engineering Lead, Project Manager

Precondition: Work sessions exist.

Trigger: Navigating to `/ai-agents/timesheet` or `/teams/timesheet`.

Table	Relevant columns currently defined	Role in this requirement
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	

Assessment: Derived: timesheet reporting comes from work_session joined to task/project/org_member and agent_event. Clarify treatment of auto-stopped sessions and reviewed time.

SRS acceptance criteria: Timesheet renders distinct color-coded columns for human time vs AI agent time.

Documents & contracts

FR-DOC-001 — Hierarchical Document Tree

SRS behaviour: The system shall provide a hierarchical document storage repository with nested folders, document counts, and breadcrumb navigation.

Actor: Document Manager, Project Manager, Org Admin

Precondition: Authenticated user with document access.

Trigger: Navigation to `/documents` or `/documents/folders`.

Table	Relevant columns currently defined	Role in this requirement
document_folder	document_folder_id (PK), organization_id, parent_folder_id, owner_user_id, name, created_at, updated_at	
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
document_share	document_share_id (PK), organization_id, document_id, user_id, access_type, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	

Assessment: Covered: document_folder parent hierarchy and document.folder link support tree. Add same-organisation parent-folder constraints and folder/document name collision rules if needed.

SRS acceptance criteria: Documents inside folders filter correctly when navigating into subfolders.

FR-DOC-002 — Document Version History

SRS behaviour: The system shall maintain an incremental version history for editable documents, tracking version number, author, modified date, change summary, and full content snapshot.

Actor: Document Manager, Product Owner

Precondition: Document exists in editable mode.

Trigger: Saving modifications in `/documents/[documentId]/edit`.

Table	Relevant columns currently defined	Role in this requirement
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
document_version	document_version_id (PK), organization_id, document_id, modified_by_user_id, stored_file_id, version_number, change_summary, content, content_sha256, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: document_version supports version snapshots and content hashes. Clarify whether every save creates a version or only published/approved/signing checkpoints; enforce immutable signed versions.

SRS acceptance criteria: Version history drawer displays all historical snapshots with timestamps and change summaries.

FR-DOC-003 — Template Variable Tags

SRS behaviour: The system shall maintain standardized document templates across legal, sales, HR, and PM categories with dynamic placeholder tags (e.g., `{{client_name}}`, `{{effective_date}}`).

Actor: Document Manager, Project Manager

Precondition: Navigation to `/documents/templates`.

Trigger: Creating a document from a template.

Table	Relevant columns currently defined	Role in this requirement
document_template	document_template_id (PK), organization_id, created_by_user_id, title, description, category, type, content, created_at, updated_at	
document_template_placeholder	document_template_placeholder_id (PK), organization_id, document_template_id, placeholder_key, label, sort_order	
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
user_favorite_template	user_favorite_template_id (PK), organization_id, user_id, document_template_id, created_at	

Assessment: Partial: template placeholders define keys/labels but no separate document_template_placeholder_value table. If placeholders are substituted into document.content at generation, no extra table is needed; if values must be editable/auditable individually, add one.

SRS acceptance criteria: Template placeholders are dynamically converted into form fields and substituted into the resulting document.

FR-DOC-004 — Contract Lifecycle States

SRS behaviour: The system shall manage contracts through an explicit lifecycle (`draft`, `pending_review`, `awaiting_signature`, `active`, `expiring`, `completed`, `expired`, `archived`), capturing counterparties, values, and dates.

Actor: Document Manager, Executive / CEO

Precondition: User possesses contract management access.

Trigger: Creating or updating a contract in `/contracts`.

Table	Relevant columns currently defined	Role in this requirement
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
document_version	document_version_id (PK), organization_id, document_id, modified_by_user_id, stored_file_id, version_number, change_summary, content, content_sha256, created_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group,	

	accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: contract status and expiry dates support lifecycle. Need explicit state transition history if the product must show who changed contract status and when; activity_log can provide a generic audit trail.

SRS acceptance criteria: Contract status chip accurately updates across review, signing, and active execution stages.

FR-DOC-005 — Contract Review Comments

SRS behaviour: The system shall allow reviewers to leave comments on contracts requesting changes or signifying preliminary approval before signing.

Actor: Reviewer, Document Manager, Legal Counsel

Precondition: Contract is in `pending_review` status.

Trigger: Submitting a review comment on `/contracts/[contractId]`.

Table	Relevant columns currently defined	Role in this requirement
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	
contract_review_comment	contract_review_comment_id (PK), organization_id, contract_id, author_user_id, text, action_requested, created_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: contract_review_comment captures reviewer comments and action requested. Clarify whether multiple reviewers/approval stages are required; current contract has only one reviewer_user_id.

SRS acceptance criteria: Review comments display author, timestamp, and requested action badges.

FR-DOC-006 — Project Cross-Linking

SRS behaviour: The system shall support associating documents and contracts to specific project records (`relatedProjectId`), linking project milestone deliverables to binding agreements.

Actor: Project Manager, Document Manager

Precondition: Project and Document exist.

Trigger: Selecting a project in document or contract settings.

Table	Relevant columns currently defined	Role in this requirement
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title,	

	description, type, status, share_scope, content, created_at, updated_at, deleted_at	
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at	

Assessment: Covered: document.project_id and contract.project_id cross-link content to projects; project_file handles SRS/test report files.

SRS acceptance criteria: Project overview renders all linked contracts and documents in its resources tab.

FR-DOC-007 — Document Favoriting

SRS behaviour: The system shall allow users to mark documents as favorites and filter repository lists by Favorites, Shared With Me, Recent, and Created by Me.

Actor: Authenticated User

Precondition: Navigation to `/documents`.

Trigger: Clicking the star icon on a document row or card.

Table	Relevant columns currently defined	Role in this requirement
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
user_favorite_document	user_favorite_document_id (PK), organization_id, user_id, document_id, created_at	
user_recent_document	user_recent_document_id (PK), organization_id, user_id, document_id, last_opened_at	

Assessment: Covered: user_favorite_document supports favorites; user_recent_document supports recent items.

SRS acceptance criteria: Starred documents appear instantly in the Favorites folder view.

FR-DOC-008 — Document PDF Download

SRS behaviour: The system shall allow users to download stored documents and contracts as rendered PDF files.

Actor: Authenticated User

Precondition: Document exists.

Trigger: Clicking \"Download PDF\" in document toolbar.

Table	Relevant columns currently defined	Role in this requirement
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content,	

	created_at, updated_at, deleted_at	
document_version	document_version_id (PK), organization_id, document_id, modified_by_user_id, stored_file_id, version_number, change_summary, content, content_sha256, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: stored_file and document_version stored_file_id support PDF download/version files. Ensure only clean scanned files can be served.

SRS acceptance criteria: Clicking download retrieves valid PDF byte stream.

Electronic signatures

FR-SIGN-001 — Envelope Builder Wizard

SRS behaviour: The system shall provide an envelope creation wizard allowing users to upload documents, add recipients with specific roles, define email messages, and configure reminder frequencies.

Actor: Any organisation user

Precondition: User is a member of the organisation. Agreements has no role-level restriction (DEC-27).

Trigger: Agreements › Get signatures, or Documents › Send for signature on a PDF document (`/signatures/upload?mode=send`). The prototype's older `/signatures/new` wizard serves text documents only and is to be retired (GAP-16).

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_file	signature_request_file_id (PK), organization_id, signature_request_id, stored_file_id, document_id, sort_order	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
copy_recipient	copy_recipient_id (PK), organization_id, signature_request_id, name, email, role, sort_order	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	
signature_field_option	signature_field_option_id (PK), organization_id, signature_field_id, label, sort_order	
signing_default	signing_default_id (PK), organization_id, expire_days, reminder_frequency, signing_order, email_subject, message, expiry_warning_days, updated_at	
signing_default_method	signing_default_method_id (PK), organization_id, signing_default_id, method	

project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: signature_request plus request_file, signer, reviewer, copy_recipient, fields/options, signing defaults and stored files support the wizard.

SRS acceptance criteria: Wizard validates document MIME types and creates draft with assigned recipient IDs.

FR-SIGN-002 — Coordinate Field Placement

SRS behaviour: The system shall render an interactive PDF canvas allowing senders to drag and drop signature, initials, date, name, text, and checkbox fields onto specific pages for designated recipients.

Actor: Document Manager, Project Manager

Precondition: Draft envelope has at least 1 document and 1 signer.

Trigger: Navigating to Step 2: Add Fields (`/envelopes/get-signatures/add-fields`).

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_file	signature_request_file_id (PK), organization_id, signature_request_id, stored_file_id, document_id, sort_order	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	
signature_field_option	signature_field_option_id (PK), organization_id, signature_field_id, label, sort_order	

stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: signature_field stores page number and percentage coordinates, signer assignment, field type and value; options are normalised.

SRS acceptance criteria: Placed fields maintain consistent relative positions across various screen zooms (50% to 400%).

FR-SIGN-003 — Send Validation Gate

SRS behaviour: Before sending, the system shall validate that every recipient assigned to sign has at least one signature/initial field placed, activating recipients according to routing order.

Actor: Sender (Document Manager / Project Manager)

Precondition: Envelope is in `draft` status.

Trigger: Clicking \"Send Envelope\".

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_file	signature_request_file_id (PK), organization_id, signature_request_id, stored_file_id, document_id, sort_order	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
signing_default	signing_default_id (PK), organization_id, expire_days, reminder_frequency, signing_order, email_subject, message, expiry_warning_days, updated_at	
company_seal	company_seal_id (PK), organization_id, created_by_user_id, image_file_id, name, description, is_active, created_at, updated_at	

Assessment: Partial: required fields, recipients, defaults and seal checks can validate sending. Persist a validation snapshot only if the exact pre-send state must be audited; otherwise event log may suffice.

SRS acceptance criteria: Unassigned signers prevent envelope sending; sequential routing emails only group 1 signers upon initial send.

FR-SIGN-004 — Token-Gated Public Signing

SRS behaviour: The system shall allow recipients to open and execute agreements via a personal signing link (`/sign/[token]`) carrying a random token of at least 128 bits, bound to one recipient and one request (NFR-022), without requiring account registration, while enforcing a login gate if the recipient's email belongs to an existing organization account.

Actor: Recipient (Signer)

Precondition: Active envelope exists with valid signing token.

Trigger: Recipient accesses `/sign/[token]`.

Table	Relevant columns currently defined	Role in this requirement
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	

Assessment: Partial: signing_token and signer/reviewer links support token-gated access. The schema has no dedicated recipient email-OTP challenge table despite signer.auth_method allowing email_otp; add signer_verification_challenge (hashed code, attempts, expires_at, consumed_at) if email OTP is required.

SRS acceptance criteria: Registered recipients cannot sign without authenticating; unregistered recipients execute via token.

FR-SIGN-005 — Responsive Signing Pad

SRS behaviour: The system shall provide an accessible modal allowing signers to adopt a signature and initials by typing in decorative script fonts, drawing on a canvas, or uploading an image.

Actor: Signer

Precondition: Signing interface is active.

Trigger: Clicking a signature or initials field.

Table	Relevant columns currently defined	Role in this requirement
saved_signature	saved_signature_id (PK), user_id, signature_file_id, initials_file_id, name, initials, source, style, updated_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	

Assessment: Partial: UI pad input can be stored via saved_signature/stored_file and signer signature fields. Need record explicit consent/signature method at the signing event and ensure uploaded/drawn assets are linked to the signer and request.

SRS acceptance criteria: Adopting a signature fills all matching fields across the document in one action.

FR-SIGN-006 — Company Seal Role Gate

SRS behaviour: The system shall allow the roles chosen on each seal (e.g., Organization Owner, Project Manager) to stamp official corporate seals on agreements

Actor: CEO / Executive, Authorized Officer

Precondition: Company seals exist in workspace with configured `allowedRoles`.

Trigger: Applying seal in signing interface.

Table	Relevant columns currently defined	Role in this requirement
company_seal	company_seal_id (PK), organization_id, created_by_user_id, image_file_id, name, description, is_active, created_at, updated_at	
company_seal_allowed_role	company_seal_allowed_role_id (PK), organization_id, company_seal_id, role_id	
company_seal_use	company_seal_use_id (PK), organization_id, company_seal_id, signature_request_id, signer_id, signature_field_id, used_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign,	

	reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: company_seal_allowed_role defines role eligibility; company_seal_use links actual use to request, signer and field. Check role at send and at execution time.

SRS acceptance criteria: Only users holding designated roles can view and apply company seals.

FR-SIGN-007 — Sequential Signing Handoff

SRS behaviour: When an active signer completes submission, the system shall record field values, advance status to signed, activate the next routing order group, or mark the envelope completed when all signers have finished.

Actor: Signer

Precondition: All required fields for the signer are populated.

Trigger: Clicking \"Finish Signing\".

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	

Assessment: Covered: signer.signing_position and signature_request.signing_order/status support sequential handoff. Ensure only the current eligible signer can act.

SRS acceptance criteria: Completing signer 1 automatically triggers invite email to signer 2. Completing the final signer on an envelope without reviewers completes it, flattens the PDF and sends completion emails. On an envelope with reviewers it moves the envelope to `awaiting_review` and sends review invitations instead.

FR-SIGN-008 — Recipient Decline Workflow

SRS behaviour: The system shall allow signers to decline executing an agreement, capturing a mandatory decline reason, marking the recipient as `declined`, and immediately voiding the envelope.

Actor: Signer

Precondition: Recipient status is `pending` or `viewed` (it is their turn).

Trigger: Clicking \"Decline\" in signing interface.

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	

Assessment: Covered: signer declined_at/reason/status and signature_request_event support decline workflow. Ensure remaining signers and reviewers are transitioned consistently.

SRS acceptance criteria: Declining an envelope prevents any subsequent signers from executing and sends notification to sender.

FR-SIGN-009 — Saved Signature Profile

SRS behaviour: The system shall maintain exactly one saved signature and initials per registered user profile, pre-filling fields automatically during self-signing and agreement signing.

Actor: Authenticated User

Precondition: Navigation to Settings $\rightarrow$ Signature.

Trigger: Saving a signature in `/settings/signature` or profile.

Table	Relevant columns currently defined	Role in this requirement
saved_signature	saved_signature_id (PK), user_id, signature_file_id, initials_file_id, name, initials, source, style, updated_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value	

Assessment: Covered: saved_signature and stored_file support one reusable personal signature profile. Define replacement/deletion rules and prevent cross-user file access.

SRS acceptance criteria: Updating signature in settings immediately updates default signature in signing workflows.

FR-SIGN-010 — PDF Flattening (`pdf-lib`)

SRS behaviour: The system shall burn collected signatures, initials, dates, and text values directly into the PDF byte stream, transforming top-left page-percentage coordinates into bottom-left PDF point coordinates.

Actor: System

Precondition: Envelope status transitions to `completed` or signed PDF download requested.

Trigger: System executes `flattenEnvelopePdf`.

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_file	signature_request_file_id (PK), organization_id, signature_request_id, stored_file_id, document_id, sort_order	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y,	

	width, height, is_required, value	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
signature_evidence	signature_evidence_id (PK), organization_id, signature_request_id, stored_file_id, kind, detail, created_at	

Assessment: Partial: final PDF and evidence files are represented by signature_request.signed_file_id, certificate_file_id and signature_evidence. PDF flattening itself is processing logic; preserve original source PDFs and final SHA-256 hash.

SRS acceptance criteria: Output PDF renders embedded signatures and text at exact coordinates matching the web placement view.

FR-SIGN-011 — Background Sweeper Daemon

SRS behaviour: The system shall execute background sweeps to send automated reminder emails to pending signers based on configured frequency and expire envelopes that have exceeded their validity lifespan.

Actor: System Background Scheduler

Precondition: Scheduler is running.

Trigger: Periodic execution (default every 15 minutes).

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Partial: timestamps/reminder frequency/expiry and outbound_email support sweeper jobs. Add a durable job/outbox/attempt record if jobs need retries, deduplication and exactly-once effects; outbound_email may be sufficient if treated as an outbox.

SRS acceptance criteria: Overdue envelopes automatically transition to `expired` and disallow further signature execution.

FR-SIGN-012 — Immutable Agreement Audit

SRS behaviour: The system shall record and display an immutable chronological audit trail for each envelope capturing created, sent, viewed, signed, declined, cancelled, expired, reminder sent, review requested, review approved, review rejected and completed events with IP addresses and actor emails.

Actor: Auditor, Document Manager, Signer

Precondition: Envelope exists.

Trigger: Any lifecycle state change or viewing `/signatures/[requestId]/audit`.

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
signature_evidence	signature_evidence_id (PK), organization_id, signature_request_id, stored_file_id, kind, detail, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: signature_request_event is append-only and hash chained; activity_log provides wider organisation audit. Ensure database privileges prohibit mutation and hash-chain verification is periodically run.

SRS acceptance criteria: Audit trail details exact timestamps for envelope dispatch, recipient viewing, and execution.

FR-SIGN-013 — Advanced Electronic Signature Assurance

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id,	

	sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at	
signature_evidence	signature_evidence_id (PK), organization_id, signature_request_id, stored_file_id, kind, detail, created_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Partial / decision-dependent: signature_evidence.kind and detail are generic. Advanced assurance may require structured certificate issuer/serial, signer identity proof, timestamp authority response, signature validation result, algorithm and revocation-check evidence. DEC-24 remains open.

FR-SIGN-014 — Reviewer Approval Before Finalisation

SRS behaviour: The system shall let the sender add one or more reviewers to a signature request. Reviewers do not sign. After every signer has signed, each reviewer checks the signed document and approves or rejects it. A request with reviewers shall be completed only when every reviewer has approved. A request without reviewers shall be completed on the last signature (FR-SIGN-007).

Actor: Sender (adds reviewers), Reviewer

Precondition: The request was sent with at least one recipient whose role is `needs_to_review`, and every signer has signed.

Trigger: The last signer finishes signing (FR-SIGN-007).

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent,	

	prev_hash, hash, created_at	
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at	

Assessment: Covered: reviewer rows and reviewer status/note/reviewed_at plus signature_request.awaiting_review support reviewer approval before finalisation.

SRS acceptance criteria: Given one signer and one reviewer, when the signer finishes, then the request is `awaiting_review` and the reviewer receives a review email. When the reviewer approves, then the request is `completed` and everyone receives the signed PDF. Given the same request, when the reviewer rejects with a reason, then the request is `declined` and the sender and the signer are emailed the reason. Given a request with no reviewer, when the last signer finishes, then it is `completed` immediately.

Team collaboration

FR-TEAM-001 — Team Roster Management

SRS behaviour: The system shall allow administrators to create named teams/squads, assign team leads, set department associations, and assign member rosters.

Actor: Users whose role has Team Management access (others see Access Denied on `/teams/new`)

Precondition: Organization users exist.

Trigger: Creating or editing a team on `/teams` or `/teams/new`.

Table	Relevant columns currently defined	Role in this requirement
team	team_id (PK), organization_id, lead_user_id, department_id, name, description, category, color, icon, status, created_at, updated_at	
team_member	team_member_id (PK), organization_id, team_id, user_id, created_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
department	department_id (PK), organization_id, name, sort_order, created_at	
team_project	team_project_id (PK), organization_id, team_id, project_id	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	

Assessment: Covered: team, team_member, department and team_project represent rosters and team/project membership.

SRS acceptance criteria: Teams display accurate member counts and aggregate projects assigned to team members.

FR-TEAM-002 — Member Capacity Hours

SRS behaviour: The system shall record individual team member weekly capacity hours (default 40h), working hours, timezones, and skills to drive workload analytics.

Actor: Team Management roles (any member); every user (their own profile)

Precondition: User record exists.

Trigger: Editing profile in `/settings/profile` or `/teams/members/[memberId]`.

Table	Relevant columns currently defined	Role in this requirement
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
time_off_request	time_off_request_id (PK), organization_id, user_id, decided_by_user_id, leave_type, start_date, end_date, duration_days, is_partial_day, reason, notes, status, decision_note, decided_at, created_at, updated_at	

Assessment: Partial: org_member working hours/capacity/availability and work_session provide capacity/workload. Consider storing effective-dated capacity history if past workload reports must use historical capacity.

SRS acceptance criteria: Member capacity hours reflect accurately in team workload balance charts.

FR-TEAM-003 — Time-Off Request Lifecycle

SRS behaviour: The system shall allow members to submit time-off requests across leave types (`vacation`, `sick`, `personal`, `parental`, `unpaid`), supporting manager approval and calendar blocking.

Actor: Team Member (Submitter); Project Manager / Lead (Approver)

Precondition: User belongs to organization.

Trigger: Submitting \"Request Time Off\" modal in `/planner`.

Table	Relevant columns currently defined	Role in this requirement
time_off_request	time_off_request_id (PK), organization_id, user_id, decided_by_user_id, leave_type, start_date, end_date, duration_days, is_partial_day, reason, notes, status, decision_note, decided_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
notification	notification_id (PK), organization_id, user_id, notification_event_id, type, title, body, link, is_read, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: time_off_request includes dates, type, duration, decision and notes. Add approver policy/backup approver only if required by workflow.

SRS acceptance criteria: Approved leave displays on planning calendar and alerts managers of reduced capacity during sprint planning.

FR-TEAM-004 — Meeting Scheduling

SRS behaviour: The system shall allow users to schedule team meetings with date, start/end time, participant lists, meeting agendas, and video conference links (e.g., Google Meet, Zoom).

Actor: Authenticated User

Precondition: Navigation to `/planner`.

Trigger: Submitting \"Schedule Meeting\" dialog.

Table	Relevant columns currently defined	Role in this requirement
meeting	meeting_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, start_at, end_at, location, meeting_link, status, created_at, updated_at	
meeting_participant	meeting_participant_id (PK), organization_id, meeting_id, user_id	
meeting_agenda_item	meeting_agenda_item_id (PK), organization_id, meeting_id, item, sort_order	
calendar_event	calendar_event_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, category, start_at, end_at, all_day, location, meeting_link, created_at, updated_at	
calendar_event_attendee	calendar_event_attendee_id (PK), organization_id, calendar_event_id, user_id	
notification	notification_id (PK), organization_id, user_id, notification_event_id, type, title, body, link, is_read, created_at	

Assessment: Covered: meeting, participants, agenda and calendar events support meeting scheduling. Clarify whether meeting and calendar_event are intentionally separate records or should share one canonical event.

SRS acceptance criteria: Scheduled meetings display on the calendar with working external video join links.

FR-TEAM-005 — Company Announcements

SRS behaviour: The system shall provide an organization announcement feed where leaders can publish prioritized announcements with rich text content.

Actor: Org Admin, CEO, Project Manager (Publisher); All Members (Readers)

Precondition: Administrative or managerial access.

Trigger: Publishing announcement on `/teams/announcements/new`.

Table	Relevant columns currently defined	Role in this requirement
announcement	announcement_id (PK), organization_id, author_user_id, title, content, priority, is_pinned, status, published_at, created_at, updated_at	
announcement_department	announcement_department_id (PK), organization_id, announcement_id, department_id	
announcement_read	announcement_read_id (PK), organization_id, announcement_id, user_id, read_at	
department	department_id (PK), organization_id, name, sort_order, created_at	
notification	notification_id (PK), organization_id, user_id, notification_event_id, type, title, body, link, is_read, created_at	

Assessment: Covered: announcement plus department targeting and read receipts supports announcements. Notification generation/delivery is a separate concern.

SRS acceptance criteria: Important announcements render with prominent visual alert styling across team dashboards.

FR-TEAM-006 — Team Activity Stream

SRS behaviour: The system shall record and display an aggregated chronological stream of team actions across task moves, comments, approvals, document creations, and signature events.

Actor: Authenticated User

Precondition: User holds activity view permissions.

Trigger: Navigation to `/activity` or `/teams/activity`.

Table	Relevant columns currently defined	Role in this requirement
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	
notification	notification_id (PK), organization_id, user_id, notification_event_id, type, title, body, link, is_read, created_at	
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
meeting	meeting_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, start_at, end_at, location, meeting_link, status, created_at, updated_at	
announcement	announcement_id (PK), organization_id, author_user_id, title, content, priority, is_pinned, status, published_at, created_at, updated_at	

Assessment: Partial: activity_log captures actions; notification is a user inbox. Define which actions appear in the feed and whether activity is filtered by team/project.

SRS acceptance criteria: Every major entity mutation automatically prepends a new event card to the activity feed.

FR-TEAM-007 — Member Directory Search

SRS behaviour: The system shall provide a member directory with search across names, job titles, departments, and specific technical skills.

Actor: Authenticated User

Precondition: Navigation to `/teams/members` or `/teams/directory`.

Trigger: Entering search query or selecting department filter.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
department	department_id (PK), organization_id, name,	

	sort_order, created_at	
team	team_id (PK), organization_id, lead_user_id, department_id, name, description, category, color, icon, status, created_at, updated_at	
team_member	team_member_id (PK), organization_id, team_id, user_id, created_at	
org_member_skill	org_member_skill_id (PK), organization_id, org_member_id, skill	

Assessment: Covered: org_member/users/department/team/skills support directory search. Add indexes for name/email/department/skill search as needed.

SRS acceptance criteria: Searching for a specific skill returns only members tagged with that technical competency.

FR-TEAM-008 — Team Access Levels and Member Management

SRS behaviour: The system shall give Team Management roles full control of the Team section and give Team Member roles a view of the organisation's members only.

Actor: Organization Owner; Team Management roles; Team Member roles

Precondition: User's role has Team Access set (Section 2.5).

Trigger: Opening Team › Analytics (`/teams`), Members (`/teams/members`) or Roles & Permissions (`/teams/roles`).

Table	Relevant columns currently defined	Role in this requirement
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
invitation_project	invitation_project_id (PK), organization_id, invitation_id, project_id	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
department	department_id (PK), organization_id, name, sort_order, created_at	
team_member	team_member_id (PK), organization_id, team_id, user_id, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: team_access role settings, org_member and team_member support access levels and membership. The schema does not explicitly distinguish a member's team-management scope from organisation-wide team management; confirm the SRS means org-wide.

SRS acceptance criteria: Owner and Project Manager can perform every Team action; Team Lead, Developer and QA Tester see only Members, read-only, and cannot reach Analytics, Roles & Permissions or any create, edit or delete action.

FR-TEAM-009 — Custom Role Creation and Editing

SRS behaviour: The system shall let the Organization Owner and Team Management roles view, create, edit and delete organisation roles, each defined by Basic Information, Dashboard Access, Project Access, My Tasks access and Team Access.

Actor: Organization Owner; Team Management roles

Precondition: User's role has Team Management access.

Trigger: Create Custom Role, or Edit on a role card, in Team › Roles & Permissions (`/teams/roles`).

Table	Relevant columns currently defined	Role in this requirement
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
role_template	role_template_id (PK), role_key, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, sort_order	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Covered: role supports custom role CRUD and four access settings; role_template supplies defaults. Protect system roles and prevent deletion while assigned or referenced by pending invitations.

SRS acceptance criteria: The Owner creates a role with My Status only, My Tasks No, Project User Access and Team Member; a user given that role lands on My Status, has no My Tasks, sees only assigned projects and has Team Member restrictions.

Reports & analytics

FR-REP-001 — Portfolio Health Overview

SRS behaviour: The system shall aggregate organization-wide project metrics, rendering distributions of project health (`on_track`, `at_risk`, `off_track`, `delayed`), overall delivery progress, and active project counts.

Actor: CEO / Executive, Project Manager

Precondition: Navigation to `/dashboard` or `/reports`.

Trigger: Page rendering.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	

project_goal	project_goal_id (PK), organization_id, project_id, goal, sort_order	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at	
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Derived: portfolio metrics are aggregations over projects, tasks, plans and work sessions. DEC-28 must determine assigned-project vs organisation-wide visibility for User Access dashboard roles.

SRS acceptance criteria: Project health chart segments correspond exactly to the count of projects in each health state.

FR-REP-002 — Time Tracking & AI Share

SRS behaviour: The system shall analyze work session records to report hours logged per contributor and compute the ratio of human engineering hours versus AI coding agent hours.

Actor: Engineering Lead, Project Manager, Executive

Precondition: Work sessions exist in state.

Trigger: Navigating to Task Time report in `/reports`.

Table	Relevant columns currently defined	Role in this requirement
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date,	

	acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at	
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	

Assessment: Derived: time tracking and AI share are calculated from work_session and agent_event/agent_token_usage. Define how manual, agent and manager-entered sessions are distinguished and reconciled.

SRS acceptance criteria: Report distinguishes time logged by Claude, Codex, Gemini, Antigravity, and human developers.

FR-REP-003 — Schedule Slippage Report

SRS behaviour: The system shall generate schedule variance reports comparing original baseline completion targets against calculated forecast dates across all projects.

Actor: Project Manager, Executive

Precondition: Projects have baseline and calculated timelines.

Trigger: Viewing Timeline report in `/reports`.

Table	Relevant columns currently defined	Role in this requirement
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at	
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at	
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at	

Assessment: Derived: schedule slippage uses baseline_change and project/task/phase dates. Define the exact baseline and variance formula.

SRS acceptance criteria: Projects with positive variance days are flagged as extended with exact slipped day counts.

FR-REP-004 — Team Workload Heatmap

SRS behaviour: The system shall calculate individual team member workload allocations against weekly capacities, highlighting over-allocated members in red.

Actor: Project Manager, Engineering Lead

Precondition: Tasks assigned with estimates; member weekly capacities configured.

Trigger: Navigating to `/teams/workload`.

Table	Relevant columns currently defined	Role in this requirement
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note	
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at	
time_off_request	time_off_request_id (PK), organization_id, user_id, decided_by_user_id, leave_type, start_date, end_date, duration_days, is_partial_day, reason, notes, status, decision_note, decided_at, created_at, updated_at	
team	team_id (PK), organization_id, lead_user_id, department_id, name, description, category, color, icon, status, created_at, updated_at	
team_member	team_member_id (PK), organization_id, team_id, user_id, created_at	

Assessment: Derived: workload heatmap uses org_member capacity, work_session, tasks and time_off_request. Historical capacity snapshots may be required for accurate past reporting.

SRS acceptance criteria: Members assigned >40 hours of estimated work in a single week trigger visual over-allocation alerts.

FR-REP-005 — Agreement Velocity Metrics

SRS behaviour: The system shall report signature request throughput, displaying counts of Sent, Viewed, Signed, Declined, Expired, and Draft agreements alongside average turnaround times.

Actor: Document Manager, Executive / CEO

Precondition: Signature requests exist.

Trigger: Navigation to `/agreements/reports`.

Table	Relevant columns currently defined	Role in this requirement
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id,	

	cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at	
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at	

Assessment: Derived: agreement velocity uses signature_request lifecycle timestamps and signer events. Define the metric start/end points and treatment of declined/cancelled/expired requests.

SRS acceptance criteria: Agreement reports render accurate counts matching the active signature requests dataset.

FR-REP-006 — System Audit Log Viewer

SRS behaviour: The system shall provide compliance auditors and administrators with a centralized, searchable log of all security and data modification events across users, tasks, and agreements.

Actor: Compliance Auditor, Org Admin

Precondition: User holds auditor or admin permissions.

Trigger: Viewing `/settings/activity` or `/reports`.

Table	Relevant columns currently defined	Role in this requirement
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at	
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	

Assessment: Partial: activity_log, signature_request_event and platform_activity_log cover most audit views. Add a unified audit query/view or event correlation convention; platform-level and tenant-level logs must remain distinct.

SRS acceptance criteria: Security events (logins, token revocations, signature completions) appear in the audit table with timestamps and actor attribution.

Settings & governance

FR-SETT-001 — Organization Branding

SRS behaviour: The system shall allow administrators to update organization display name, slug, logo URL, industry category, and subscription plan tier (`starter`, `pro`, `enterprise`).

Actor: Organization Owner

Precondition: User is the Organization Owner. The Preferences tab is hidden from all other roles and `/settings/preferences` shows Access Denied ("Owner Authority Required"); other Settings pages keep their existing rules.

Trigger: Updating the organisation form in `/settings/preferences`.

Table	Relevant columns currently defined	Role in this requirement
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: organisation branding fields and logo_file_id are present.

SRS acceptance criteria: Changing organization name in settings updates top navigation breadcrumbs and sidebar workspace labels.

FR-SETT-002 — Member Role Administration

SRS behaviour: The system shall permit the Owner and Team Management roles to change a member's organisation role, update assigned projects, job titles and departments, or deactivate or remove departing members.

Actor: Organization Owner; Team Management roles

Precondition: User's role has Team Management access (Team › Members, or the Users & Roles tab in Settings).

Trigger: Modifying user attributes in user management table.

Table	Relevant columns currently defined	Role in this requirement
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Partial: org_member.role_id and invitation.role_id support role assignment; suspension/removal fields exist. Ensure changes revoke sessions or active agent tokens according to policy.

SRS acceptance criteria: Modifying a member's role instantly updates their permitted navigation groups and action gates.

FR-SETT-003 — Company Seal Configuration

SRS behaviour: The system shall allow administrators to upload corporate seal stamp images (PNG/SVG data URLs), configure active status, and designate which roles (`allowedRoles`) are authorized to stamp them on agreements.

Actor: Org Admin, CEO

Precondition: Admin permissions.

Trigger: Adding or editing a seal in Agreements $\rightarrow$ Settings $\rightarrow$ Company Seals.

Table	Relevant columns currently defined	Role in this requirement
company_seal	company_seal_id (PK), organization_id, created_by_user_id, image_file_id, name, description, is_active, created_at, updated_at	
company_seal_allowed_role	company_seal_allowed_role_id (PK), organization_id, company_seal_id, role_id	
company_seal_use	company_seal_use_id (PK), organization_id, company_seal_id, signature_request_id, signer_id, signature_field_id, used_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at	

Assessment: Covered: company seal, allowed roles and usage history are modelled.

SRS acceptance criteria: Uploaded seals appear in signing pad only for users possessing authorized roles.

FR-SETT-004 — Global Signing Defaults

SRS behaviour: The system shall allow administrators to define global defaults for new signature requests, including default expiration days (default 120), reminder frequency, default email subject/message, and permitted signature methods.

Actor: Org Admin, Document Manager

Precondition: Admin access.

Trigger: Saving settings in `/signatures/settings`.

Table	Relevant columns currently defined	Role in this requirement
signing_default	signing_default_id (PK), organization_id, expire_days, reminder_frequency, signing_order, email_subject, message, expiry_warning_days, updated_at	
signing_default_method	signing_default_method_id (PK), organization_id, signing_default_id, method	

Assessment: Covered: signing_default and signing_default_method represent organisation signing defaults.

SRS acceptance criteria: New envelope drafts automatically populate with configured default message and expiration days.

FR-SETT-005 — Notification Preferences

SRS behaviour: The system shall allow users to customize notification delivery preferences, independently toggling in-app alerts, email notifications, task assignments, deadline reminders, and approval requests.

Actor: Authenticated User

Precondition: Navigation to Settings $\rightarrow$ Notifications.

Trigger: Toggling switches in `/settings/notifications`.

Table	Relevant columns currently defined	Role in this requirement
user_preference	user_preference_id (PK), user_id, theme, default_calendar_view, default_planning_view, in_app_notifications, email_notifications, desktop_notifications, weekly_digest, updated_at	
notification_event	notification_event_id (PK), event_key, name, category, default_enabled, sort_order	
user_notification_setting	user_notification_setting_id (PK), user_id, notification_event_id, is_enabled	

Assessment: Covered: user_preference and user_notification_setting support personal notification preferences. Confirm whether per-organisation overrides are needed; current settings are global per person.

SRS acceptance criteria: Toggling off task assignment emails suppresses email dispatch when a task is assigned to that user.

FR-SETT-006 — Demo State Reset Utility

SRS behaviour: The system shall provide an administrative reset utility allowing evaluators to wipe all custom `localStorage` state and restore pristine factory mock data.

Actor: Evaluator / Administrator

Precondition: Prototype execution.

Trigger: Clicking \"Reset Demo Data\" in sidebar footer or `/prototype-home`.

Table	Relevant columns currently defined	Role in this requirement
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at	
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at	
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at	
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status, title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at	
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at	
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group,	

	accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at	
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at	
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at	
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at	

Assessment: Prototype-only: demo reset must never operate on production data. For test environments, reset should be an isolated seeded-fixture operation with audit and safeguards.

SRS acceptance criteria: Resetting clears all browser-created records and restores standard Acme Digital mock data.

5. Cross-requirement relationship walkthroughs

5.1 Invitation → membership → role → project access

organization_invitation is the platform-level Owner invitation. On acceptance it resolves/creates organization and links accepted_by_user_id to organization.owner_user_id. The Owner is inserted into org_member with the Organization Owner role. For member onboarding, invitation points to role and optional department; invitation_project links the invite to each assigned project. Acceptance creates/links users, org_member and project_member rows. org_member.role_id points to role; role.project_access determines whether project_member scope applies.

Relationship	Join / rule
Owner invitation	organization_invitation.organization_id → organization.organization_id; accepted_by_user_id → users.user_id; organization.owner_user_id → users.user_id
Organisation membership	org_member.organization_id + user_id; org_member.role_id → role.role_id
Role template	role.role_template_id → role_template.role_template_id; template copied into tenant role rows
Member invitation	invitation.role_id → role.role_id; invitation_project.invitation_id → invitation.invitation_id
Assigned project access	invitation_project.project_id → project.project_id; acceptance copies to project_member(project_id,user_id)
Project visibility	role.project_access='management' gives all projects; 'user' requires project_member row

5.2 Project → board → task → work history

project owns boards, plan levels and tasks. board_column belongs to board; task references project, board and board_column. Task hierarchy is represented by parent_task_id; planning links use the lowest configured level (phase_id, epic_id or story_id). Each move appends task_status_change. Time is recorded separately in work_session and linked to task, user and board.

Entity path	Purpose
project → board → board_column	Board configuration and workflow stages
project → phase → epic → story → task	Optional planning hierarchy; task links to the lowest enabled level
task → parent_task_id → task	Subtask tree
task → task_status_change	Immutable movement history across boards/columns/status

task → work_session ← users	Time tracking and timesheet source
task → task_comment / task_checklist_item / task_attachment / task_tag / task_link	Task detail tabs and supporting records

5.3 Document → contract → signature request → evidence

document is the editable/library record. document_version snapshots content or a PDF version. contract.document_id points to contract text; contract.project_id associates it with project. signature_request may point to document source files through signature_request_file.document_id, and may link contract_id. Each signature_request has signer, reviewer and copy_recipient rows; signature_field assigns fields to a signer on a request file. signing_token grants narrow recipient access. On completion, signed_file_id/certificate_file_id and signature_evidence point to stored_file.

Entity path	Purpose
document_folder → document	Library hierarchy
document → document_version → stored_file	Version snapshots and file versions
document → document_share / document_comment / document_approval	Sharing, collaboration and approval
document_template → document_template_placeholder → document	Template structure and generated content
contract → document + project	Contract metadata and associated contract text
signature_request → signature_request_file → signer/reviewer/copy_recipient	Signing envelope and recipient routing
signature_request_file → signature_field → signature_field_option	Field placement and field options
signer/reviewer → signing_token	Public recipient access
signature_request → signature_request_event / signature_evidence / outbound_email	Audit evidence, assurance files and delivery history

6. Consolidated gaps and decisions

ID	Priority	Gap / decision	Finding and recommendation
G-01	High	Signing email OTP	signer.auth_method includes email_otp, but no recipient OTP challenge table exists. Add signer_verification_challenge (or equivalent) with hashed code, attempts, expiry, consumed_at, created_at, signer_id and organization_id.
G-02	High	Advanced signature assurance	signature_evidence.detail is unstructured text. DEC-24 remains open. Define structured certificate, identity proofing, timestamp and validation evidence before claiming advanced assurance.
G-03	High	Dashboard scope	DEC-28 is open. Decide whether Admin Dashboard metrics for Project User Access roles are organisation-wide or assigned-project-only. Enforce in query layer.
G-04	High	Tenant-safe foreign keys	Schema describes composite tenant FKs, but implementation must define unique (organization_id, id) on every tenant table and composite FK constraints consistently.
G-05	High	Role changes and revocation	Specify effects of role change, member suspension/removal, org deactivation and Owner transfer on sessions, agent tokens, project memberships, pending invitations and signing authority.
G-06	Medium	Invitation atomicity	Acceptance must be one transaction; validate token, expiry, email, org status, role ownership and project ownership, then create membership/project links and mark accepted.
G-07	Medium	Onboarding completion	No explicit organization setup completion marker. Add

			organization.onboarding_completed_at or derive completion from required fields; choose explicitly.
G-08	Medium	Task watcher/mentions	No task watcher/follower or mention entity. Add task_watcher/task_mention only if My Tasks, notifications or collaboration requirements include watching/mentions.
G-09	Medium	Baseline snapshots	baseline_change captures project-level old/new dates. If timeline needs named snapshots or per-phase/task baselines, add baseline and baseline_item tables or generalise resource_type/resource_id.
G-10	Medium	Calendar recurrence	No recurrence rule or external calendar sync cursor/event identifier. Add only if recurring events or two-way sync are in scope.
G-11	Medium	Comment revision history	task_comment/document_comment have updated_at but no revision history. Decide whether edited comments need audit reconstruction.
G-12	Medium	AI provider credentials	agent_provider has configured flags but no secret reference. Store credentials in a secret manager and persist only secret reference/provider connection metadata.
G-13	Medium	MCP request correlation	agent_event lacks request/correlation ID, protocol version and latency fields. Add if support/debugging requires request-level traceability.
G-14	Medium	Outbox/retry semantics	outbound_email may serve as an outbox, but needs retry_count, next_attempt_at, locked_at/worker status or a separate job table if reliable retry/deduplication is required.
G-15	Medium	Team meeting/calendar duplication	meeting and calendar_event can represent the same event. Define canonical source and synchronisation rule to prevent duplicate calendar entries.
G-16	Medium	Contract state history	contract status changes are not modelled as a dedicated history table. activity_log may suffice; define required audit detail and immutability.
G-17	Medium	Document template values	Template placeholders exist but individual placeholder values are not persisted. Decide whether values are materialised in document.content or require document_placeholder_value.
G-18	Medium	Organisation feature flags	organization_feature exists, but define which FR modules are gated and whether feature checks apply to API and background jobs.
G-19	Medium	Stored file lifecycle	stored_file has scan status but no explicit quarantine/deletion lifecycle, storage provider, checksum algorithm version or retention/legal-hold relation for non-signature files.
G-20	Medium	Notifications delivery	notification is an inbox record; outbound_email is email history. Define push/desktop delivery attempts, dedupe key and read/unread behaviour.
G-21	Medium	Phase duration	phase has dates but no duration_unit/estimated duration although project/epic/story/task support duration planning. Confirm intentional calculation.
G-22	Medium	Capacity history	org_member capacity is current-state only. Historical workload heatmaps may require effective-dated capacity

			records.
G-23	Medium	Signature consent evidence	signer stores consent_text_version and consented_at, but ensure the exact consent text/version is retained immutably and linked to the signed event.
G-24	Medium	Signature validation lifecycle	Define how cryptographic validation, certificate revocation, timestamp validation and long-term verification results are stored and refreshed.
G-25	Low	Indexes/full text	Define indexes for project/task/document search, recipient email lookups, event timelines and dashboard aggregations; tenant indexes must lead with organization_id.
G-26	Low	Unique constraints	Add/check unique (organization_id, role_id), unique project membership, invitation role/project same-tenant, document folder names and cross-table constraints where applicable.

7. Schema strengths worth preserving

- Strong tenant isolation concept: organization_id on every tenant table, composite tenant FKs, RLS and tenant-first indexes.
- Good separation of global identity from tenant membership: users is global while org_member captures organisation-specific role/status.
- Flexible role model aligned to Round 3: role_template defaults copied into organisation-specific role records; role permissions are decomposed into four settings.
- Good task history design: task_status_change and sprint_task preserve movement and sprint scope history rather than overwriting everything.
- Work tracking is separated from tasks: work_session supports manual/agent source, auto-stop review and overlap prevention.
- Good document/signature normalisation: stored_file, document_version, signature_request_file, signer, reviewer, signature_field and event/evidence tables are distinct.
- Security-sensitive tokens are represented as hashes; signature and agent events support append-only audit chains.
- AI analysis preserves source and edits: srs_analysis_run plus original_* and edited candidate fields support human review.

8. Implementation checklist before migrations

- ☐ Resolve DEC-28 dashboard data visibility.
- ☐ Confirm email OTP versus invitation-as-verification rules and add signer email OTP challenge persistence if email_otp is supported.
- ☐ Resolve DEC-24 identity proofing and cryptographic signature evidence requirements.
- ☐ Define role-change/member-removal effects on sessions, tokens, project assignments and signing requests.
- ☐ Write composite tenant unique keys and composite foreign-key constraints for every tenant-to-tenant relationship.
- ☐ Define transaction boundaries for invitation acceptance, project/task moves, signature send/complete, and SRS candidate batch commit.
- ☐ Define audit event catalogue and append-only database privileges.
- ☐ Define object-storage upload, malware scan, quarantine, signed-file retention and legal-hold lifecycle.
- ☐ Define background job/outbox retry, deduplication and failure recovery.

- ☐ Define all derived metric formulas: project health, phase progress, schedule variance, workload, agreement velocity and dashboard scope.
- ☐ Define indexes based on actual query patterns and ensure tenant-leading indexes.
- ☐ Create migration tests that attempt cross-tenant FK writes and RLS bypass; they must fail.
- ☐ Create end-to-end tests for invitation → role → project access; task move → history; SRS analysis → candidate review → task creation; signing → audit/evidence → final PDF.

9. Conclusion

The current schema is a credible and extensive starting point for the Round 3 FlashSign requirements. It is not necessary to add a separate table for every FR: many requirements are views, calculations, state transitions or workflows over shared domain entities. The important next step is to close the high-priority decisions and enforce the relationships and transactional rules identified above.

The most urgent items are recipient email OTP persistence (if supported), advanced-signature evidence design, DEC-28 dashboard scope, composite tenant-safe foreign keys, and the lifecycle consequences of changing roles or membership. Once these are resolved, the schema can be translated into migrations and tested against the SRS acceptance criteria.

Appendix A. Current schema table inventory

This inventory lists all table names and columns detected in the supplied schema.txt. Types and detailed constraints remain in the original schema file.

Table	Columns
platform_admin	platform_admin_id (PK), user_id, granted_by_user_id, created_at, revoked_at
organization_invitation	organization_invitation_id (PK), organization_id, invited_by_user_id, accepted_by_user_id, organization_name, owner_name, owner_email, plan, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at
platform_activity_log	platform_activity_log_id (PK), actor_user_id, organization_id, action, target_type, target_id, detail, ip_address, user_agent, created_at
organization	organization_id (PK), owner_user_id, created_by_user_id, logo_file_id, name, slug, description, type, industry, company_size, website, accent_color, plan, status, timezone, working_hours_start, working_hours_end, week_start_day, workload_threshold_hours, created_at, updated_at, deleted_at
organization_feature	organization_feature_id (PK), organization_id, feature_key, is_enabled, updated_by_user_id, updated_at
support_access_grant	support_access_grant_id (PK), organization_id, platform_admin_id, approved_by_user_id, reason, starts_at, expires_at, revoked_at, created_at
users	user_id (PK), avatar_file_id, email, password_hash, name, phone, address, bio, company, timezone, avatar_color, is_active, email_verified_at, last_login_at, anonymized_at, created_at, updated_at
session	session_id (PK), user_id, active_organization_id, token_hash, remember_me, expires_at, last_seen_at, revoked_at, ip_address, user_agent, created_at
verification_code	verification_code_id (PK), user_id, purpose, code_hash, attempts, expires_at, consumed_at, created_at
user_preference	user_preference_id (PK), user_id, theme, default_calendar_view, default_planning_view, in_app_notifications, email_notifications, desktop_notifications, weekly_digest, updated_at
notification_event	notification_event_id (PK), event_key, name, category, default_enabled, sort_order
user_notification_setting	user_notification_setting_id (PK), user_id, notification_event_id, is_enabled
role_template	role_template_id (PK), role_key, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, sort_order
role	role_id (PK), organization_id, role_template_id, created_by_user_id, name, description, is_system, admin_dashboard, my_status, project_access, my_tasks_access, team_access, created_at, updated_at
org_member	org_member_id (PK), organization_id, user_id, role_id, department_id, status,

	title, working_hours_start, working_hours_end, weekly_capacity, availability, joined_at, suspended_at, removed_at, created_at, updated_at
org_member_skill	org_member_skill_id (PK), organization_id, org_member_id, skill
department	department_id (PK), organization_id, name, sort_order, created_at
invitation	invitation_id (PK), organization_id, role_id, department_id, invited_by_user_id, accepted_by_user_id, email, name, title, message, token_hash, status, sent_at, expires_at, accepted_at, revoked_at, created_at, updated_at
invitation_project	invitation_project_id (PK), organization_id, invitation_id, project_id
team	team_id (PK), organization_id, lead_user_id, department_id, name, description, category, color, icon, status, created_at, updated_at
team_member	team_member_id (PK), organization_id, team_id, user_id, created_at
team_goal	team_goal_id (PK), organization_id, team_id, goal, sort_order
stored_file	stored_file_id (PK), organization_id, uploaded_by_user_id, storage_key, file_name, mime_type, size_bytes, page_count, sha256, scan_status, scanned_at, created_at
activity_log	activity_log_id (PK), organization_id, actor_user_id, project_id, category, action, resource_type, resource_id, detail, ip_address, user_agent, prev_hash, hash, created_at
notification	notification_id (PK), organization_id, user_id, notification_event_id, type, title, body, link, is_read, created_at
integration	integration_id (PK), organization_id, connected_by_user_id, name, category, status, description, connected_at, created_at, updated_at
project	project_id (PK), organization_id, owner_user_id, name, project_key, description, template, status, health_mode, health, priority, project_group, accent_color, start_date, end_date, original_start_date, original_end_date, duration_unit, uses_phases, uses_epics, uses_stories, uses_subtasks, uses_backlog, allow_move, next_task_number, archived_at, created_at, updated_at, deleted_at
project_member	project_member_id (PK), organization_id, project_id, user_id, assigned_by_user_id, created_at
team_project	team_project_id (PK), organization_id, team_id, project_id
project_tag	project_tag_id (PK), organization_id, project_id, tag
project_goal	project_goal_id (PK), organization_id, project_id, goal, sort_order
user_favorite_project	user_favorite_project_id (PK), organization_id, user_id, project_id, created_at
user_recent_project	user_recent_project_id (PK), organization_id, user_id, project_id, last_opened_at
baseline_change	baseline_change_id (PK), organization_id, project_id, changed_by_user_id, reason, old_start_date, old_end_date, new_start_date, new_end_date, created_at
board	board_id (PK), organization_id, project_id, name, is_main, mirrors_main_board, sort_order, created_at
board_column	board_column_id (PK), organization_id, board_id, name, color, status, sort_order
phase	phase_id (PK), organization_id, project_id, name, description, start_date, end_date, created_at, updated_at
epic	epic_id (PK), organization_id, project_id, phase_id, name, description, color, start_date, end_date, duration_unit, created_at, updated_at
story	story_id (PK), organization_id, project_id, epic_id, phase_id, title, description, acceptance_criteria, priority, start_date, end_date, duration_unit, created_at, updated_at
sprint	sprint_id (PK), organization_id, project_id, name, goal, status, start_date, end_date, capacity, capacity_unit, started_at, completed_at, created_at, updated_at
sprint_task	sprint_task_id (PK), organization_id, sprint_id, task_id, added_at, removed_at, outcome
task	organization_id, project_id, board_id, board_column_id, parent_task_id, sprint_id, phase_id, epic_id, story_id, assignee_user_id, reporter_user_id, source_project_file_id, task_key, title, description, type, status, priority, rank, estimated_duration, duration_unit, effort_minutes, story_points, start_date, due_date, acceptance_criteria, source_section, is_srs_generated, next_subtask_number, created_at, updated_at, deleted_at
task_link	task_link_id (PK), organization_id, task_id, linked_task_id, link_type, created_by_user_id, created_at
task_tag	task_tag_id (PK), organization_id, task_id, tag
task_comment	task_comment_id (PK), organization_id, task_id, author_user_id, body, via, created_at, updated_at

task_checklist_item	task_checklist_item_id (PK), organization_id, task_id, text, is_completed, sort_order
task_attachment	task_attachment_id (PK), organization_id, task_id, stored_file_id, created_at
task_status_change	task_status_change_id (PK), organization_id, task_id, changed_by_user_id, from_board_id, to_board_id, from_board_column_id, to_board_column_id, from_status, to_status, via, changed_at
work_session	work_session_id (PK), organization_id, task_id, user_id, logged_by_user_id, board_id, source, started_at, ended_at, last_activity_at, end_reason, needs_review, reviewed_by_user_id, reviewed_at, note
calendar_event	calendar_event_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, category, start_at, end_at, all_day, location, meeting_link, created_at, updated_at
calendar_event_attendee	calendar_event_attendee_id (PK), organization_id, calendar_event_id, user_id
meeting	meeting_id (PK), organization_id, organizer_user_id, team_id, project_id, title, description, start_at, end_at, location, meeting_link, status, created_at, updated_at
meeting_participant	meeting_participant_id (PK), organization_id, meeting_id, user_id
meeting_agenda_item	meeting_agenda_item_id (PK), organization_id, meeting_id, item, sort_order
time_off_request	time_off_request_id (PK), organization_id, user_id, decided_by_user_id, leave_type, start_date, end_date, duration_days, is_partial_day, reason, notes, status, decision_note, decided_at, created_at, updated_at
announcement	announcement_id (PK), organization_id, author_user_id, title, content, priority, is_pinned, status, published_at, created_at, updated_at
announcement_department	announcement_department_id (PK), organization_id, announcement_id, department_id
announcement_read	announcement_read_id (PK), organization_id, announcement_id, user_id, read_at
document_folder	document_folder_id (PK), organization_id, parent_folder_id, owner_user_id, name, created_at, updated_at
document	document_id (PK), organization_id, document_folder_id, owner_user_id, project_id, document_template_id, pdf_file_id, title, description, type, status, share_scope, content, created_at, updated_at, deleted_at
document_version	document_version_id (PK), organization_id, document_id, modified_by_user_id, stored_file_id, version_number, change_summary, content, content_sha256, created_at
document_comment	document_comment_id (PK), organization_id, document_id, author_user_id, resolved_by_user_id, text, status, resolved_at, created_at, updated_at
document_share	document_share_id (PK), organization_id, document_id, user_id, access_type, created_at
user_favorite_document	user_favorite_document_id (PK), organization_id, user_id, document_id, created_at
user_recent_document	user_recent_document_id (PK), organization_id, user_id, document_id, last_opened_at
document_approval	document_approval_id (PK), organization_id, document_id, document_version_id, requested_by_user_id, assigned_to_user_id, status, comment, created_at, resolved_at
document_template	document_template_id (PK), organization_id, created_by_user_id, title, description, category, type, content, created_at, updated_at
document_template_placeholder	document_template_placeholder_id (PK), organization_id, document_template_id, placeholder_key, label, sort_order
user_favorite_template	user_favorite_template_id (PK), organization_id, user_id, document_template_id, created_at
project_file	project_file_id (PK), organization_id, project_id, stored_file_id, name, category, analysis_status, extracted_text, word_count, analyzed_at, created_at
contract	contract_id (PK), organization_id, owner_user_id, reviewer_user_id, project_id, document_id, title, contract_type, counterparty, effective_date, expiry_date, renewal_type, contract_value, currency, payment_terms, special_conditions, description, status, created_at, updated_at, deleted_at
contract_review_comment	contract_review_comment_id (PK), organization_id, contract_id, author_user_id, text, action_requested, created_at
signature_request	signature_request_id (PK), organization_id, sender_user_id, project_id, phase_id, contract_id, signed_file_id, certificate_file_id, cancelled_by_user_id, title, email_subject, message, signing_order, status, is_self_sign, reminder_frequency, expire_days, draft_step, expires_at, last_reminder_at, sent_at, completed_at, cancelled_at, retain_until, legal_hold, created_at, updated_at, deleted_at, purged_at
signature_request_file	signature_request_file_id (PK), organization_id, signature_request_id,

	stored_file_id, document_id, sort_order
signer	signer_id (PK), organization_id, signature_request_id, user_id, signature_file_id, initials_file_id, name, email, role_label, recipient_type, signing_position, status, is_required, color, signature_method, auth_method, consent_text_version, consented_at, viewed_at, signed_at, signed_ip, signed_user_agent, declined_at, decline_reason
copy_recipient	copy_recipient_id (PK), organization_id, signature_request_id, name, email, role, sort_order
reviewer	reviewer_id (PK), organization_id, signature_request_id, user_id, name, email, review_order, status, note, reviewed_at
signature_field	signature_field_id (PK), organization_id, signature_request_file_id, signer_id, value_file_id, field_type, page_number, position_x, position_y, width, height, is_required, value
signature_field_option	signature_field_option_id (PK), organization_id, signature_field_id, label, sort_order
signing_token	signing_token_id (PK), organization_id, signer_id, reviewer_id, token_hash, revoked_at, created_at
signature_request_event	signature_request_event_id (PK), organization_id, signature_request_id, actor_user_id, signer_id, reviewer_id, event_name, category, description, actor_name, actor_email, ip_address, user_agent, prev_hash, hash, created_at
signature_evidence	signature_evidence_id (PK), organization_id, signature_request_id, stored_file_id, kind, detail, created_at
outbound_email	outbound_email_id (PK), organization_id, signature_request_id, signer_id, reviewer_id, copy_recipient_id, user_id, kind, to_name, to_email, subject, body, with_signed_pdf, status, provider_message_id, error, sent_at, delivered_at, bounced_at, created_at
saved_signature	saved_signature_id (PK), user_id, signature_file_id, initials_file_id, name, initials, source, style, updated_at
signing_default	signing_default_id (PK), organization_id, expire_days, reminder_frequency, signing_order, email_subject, message, expiry_warning_days, updated_at
signing_default_method	signing_default_method_id (PK), organization_id, signing_default_id, method
company_seal	company_seal_id (PK), organization_id, created_by_user_id, image_file_id, name, description, is_active, created_at, updated_at
company_seal_allowed_role	company_seal_allowed_role_id (PK), organization_id, company_seal_id, role_id
company_seal_use	company_seal_use_id (PK), organization_id, company_seal_id, signature_request_id, signer_id, signature_field_id, used_at
agent_access_token	agent_access_token_id (PK), organization_id, user_id, revoked_by_user_id, name, token_hash, last_four, expires_at, revoked_at, created_at
agent_token_usage	agent_token_usage_id (PK), organization_id, agent_access_token_id, agent_client, last_used_at
agent_event	agent_event_id (PK), organization_id, user_id, agent_access_token_id, task_id, project_file_id, agent, action, outcome, error_code, detail, created_at
agent_setting	agent_setting_id (PK), organization_id, is_enabled, idle_timeout_minutes, day_cutoff, allow_claim_unassigned, allow_server_ai_on_blocked, allow_llm_processing, updated_at
agent_setting_allowed_agent	agent_setting_allowed_agent_id (PK), organization_id, agent_setting_id, agent_client, is_allowed
agent_blocked_project	agent_blocked_project_id (PK), organization_id, agent_setting_id, project_id
agent_provider	agent_provider_id (PK), organization_id, name, provider, model, is_configured, is_enabled, color
agent_provider_scope	agent_provider_scope_id (PK), organization_id, agent_provider_id, scope
srs_analysis_run	srs_analysis_run_id (PK), organization_id, project_file_id, requested_by_user_id, file_sha256, mode, agent_client, provider, model, status, error, started_at, finished_at
srs_task_candidate	srs_task_candidate_id (PK), organization_id, srs_analysis_run_id, created_task_id, reviewed_by_user_id, title, original_title, description, original_description, acceptance_criteria, original_acceptance_criteria, priority, original_priority, type, original_type, story_points, original_story_points, source_section, source_page, suggested_parent, decision, rejection_reason, created_at, reviewed_at
srs_task_candidate_tag	srs_task_candidate_tag_id (PK), organization_id, srs_task_candidate_id, tag

Appendix B. Functional requirement index

Requirement ID	SRS requirement title	Mapped tables
FR-AUTH-001	User Registration Form	users, organization, organization_invitation, invitation, invitation_project, org_member, role, session, verification_code, outbound_email, activity_log
FR-AUTH-002	Email OTP Verification	users, verification_code, session, outbound_email, platform_activity_log
FR-AUTH-003	4-Step Org Onboarding	organization, users, org_member, role, role_template, department, user_preference, organization_feature, stored_file, activity_log
FR-AUTH-004	Tokenized Team Invites	users, organization, organization_invitation, invitation, invitation_project, org_member, role, session, verification_code, outbound_email, activity_log
FR-AUTH-005	Login & JWT Session	users, session, org_member, organization, platform_admin, platform_activity_log
FR-AUTH-006	Password Reset via OTP	users, verification_code, session, outbound_email, platform_activity_log
FR-AUTH-007	Permission-Based Navigation & Route Protection	users, session, organization, platform_admin, org_member, role, project_member, activity_log
FR-AUTH-008	Demo Accounts and Persona Switcher (Prototype)	users, session, organization, org_member, role, project, project_member, platform_activity_log, activity_log
FR-AUTH-009	Super Admin Dashboard & Organization Management	platform_admin, users, organization, organization_invitation, platform_activity_log, outbound_email, stored_file
FR-AUTH-010	Dashboard Access & Switching	users, session, organization, platform_admin, org_member, role, project_member, activity_log
FR-PROJ-001	Project Creation	project, users, org_member, project_member, board, board_column, project_goal, project_tag, activity_log
FR-PROJ-002	Project Overview Dashboard	project, project_member, project_goal, phase, epic, story, task, board, board_column, work_session, baseline_change, activity_log
FR-PROJ-003	Dynamic Project Health	project, task, board_column, phase, epic, story, baseline_change, work_session
FR-PROJ-004	Custom Board Columns	project, board, board_column, task, task_status_change, activity_log
FR-PROJ-005	Multi-Board Support	project, board, board_column, task, task_status_change, activity_log
FR-PROJ-006	Project Archival & Soft Delete	project, project_member, task, document, contract, signature_request, activity_log
FR-PROJ-007	Phase $\rightarrow$ Epic $\rightarrow$ Story Plan	project, phase, epic, story, task, project_goal
FR-PROJ-008	Recent Projects History	user_recent_project, user_favorite_project, project, project_member
FR-PROJ-009	Portfolio Search & Filter	project, project_member, project_tag, org_member
FR-PROJ-010	Project Access Scope	project, project_member, org_member, role, invitation, invitation_project
FR-TASK-001	Task Key Generation	project, task
FR-TASK-002	Kanban Drag-and-Drop	task, project, board, board_column, task_status_change, activity_log
FR-TASK-003	Subtask Schedule Derivation	task, project, phase, epic, story
FR-TASK-004	Container Timer Suppression	task, parent_task_id, work_session
FR-TASK-005	Guarded Move Dialog	task, project, board, board_column, task_status_change, activity_log
FR-TASK-006	Task Detail Drawer Tabs	task, task_comment, task_checklist_item, task_attachment, task_link, task_tag, task_status_change, work_session, stored_file
FR-TASK-007	Schedule Overrun Warning	task, project, phase, epic, story
FR-TASK-008	Multi-Board Task Journey	task, project, board, board_column, task_status_change, activity_log
FR-TASK-009	Task Checklists	task, task_checklist_item
FR-TASK-010	Timestamped Task Comments	task, task_comment, users
FR-TASK-011	Task Multi-Filter & Search	task, task_tag, project, project_member, org_member
FR-TASK-012	Personal \"My Tasks\" Queue	task, project, project_member, org_member, role, work_session
FR-PLAN-001	Forward Schedule Engine	project, phase, epic, story, task
FR-PLAN-002	Baseline History Versioning	project, baseline_change, activity_log
FR-PLAN-003	Sprint Planning & Lifecycle	project, sprint, sprint_task, task, org_member, work_session
FR-PLAN-004	Unified Planning Calendar	calendar_event, calendar_event_attendee, task, phase, meeting, meeting_participant, time_off_request, org_member
FR-PLAN-005	Milestone Progress Rollup	phase, epic, story, task
FR-PLAN-006	Duration Unit Normalization	project, phase, epic, story, task
FR-PLAN-007	Timeline Variance Chart	project, baseline_change, phase, task
FR-SRS-001	Project File Categorization	project, project_file, stored_file
FR-SRS-002	Multi-Stage SRS Analysis	project, project_file, stored_file, srs_analysis_run, srs_task_candidate, srs_task_candidate_tag
FR-SRS-003	SRS Review & Audit Modal	project_file, srs_analysis_run, srs_task_candidate, srs_task_candidate_tag, users, activity_log
FR-SRS-004	In-Review Task Refinement	project_file, srs_analysis_run, srs_task_candidate, srs_task_candidate_tag, users, activity_log

FR-SRS-005	Candidate Task Rejection	project_file, srs_analysis_run, srs_task_candidate, srs_task_candidate_tag, users, activity_log
FR-SRS-006	Batch Task Ingestion	srs_analysis_run, srs_task_candidate, srs_task_candidate_tag, task, project, project_file, activity_log
FR-AGENT-001	Multi-Agent Configuration	agent_provider, agent_provider_scope, agent_setting, agent_setting_allowed_agent, organization
FR-AGENT-002	Personal Access Token Gen	agent_access_token, agent_token_usage, users, org_member, activity_log
FR-AGENT-003	Agent Token Revocation	agent_access_token, agent_token_usage, users, org_member, activity_log
FR-AGENT-004	Live Work Session Timer	work_session, task, project, agent_access_token, agent_event, agent_setting, activity_log
FR-AGENT-005	Idle Timeout & Auto-Stops	work_session, task, project, agent_access_token, agent_event, agent_setting, activity_log
FR-AGENT-006	MCP Tool Execution	agent_access_token, agent_token_usage, agent_event, task, project, work_session, task_comment, task_status_change, project_file, srs_analysis_run, srs_task_candidate
FR-AGENT-007	Agent Telemetry Feed	agent_event, agent_access_token, agent_token_usage, activity_log
FR-AGENT-008	AI Privacy Blocked Projects	agent_setting, agent_blocked_project, agent_event, project, project_file, srs_analysis_run
FR-AGENT-009	Agent Timesheet Reporting	work_session, task, project, org_member, agent_event
FR-DOC-001	Hierarchical Document Tree	document_folder, document, document_share, stored_file, project
FR-DOC-002	Document Version History	document, document_version, stored_file, users, activity_log
FR-DOC-003	Template Variable Tags	document_template, document_template_placeholder, document, user_favorite_template
FR-DOC-004	Contract Lifecycle States	contract, document, document_version, project, activity_log
FR-DOC-005	Contract Review Comments	contract, contract_review_comment, users, activity_log
FR-DOC-006	Project Cross-Linking	document, contract, project, project_file
FR-DOC-007	Document Favoriting	document, user_favorite_document, user_recent_document
FR-DOC-008	Document PDF Download	document, document_version, stored_file
FR-SIGN-001	Envelope Builder Wizard	signature_request, signature_request_file, signer, reviewer, copy_recipient, signature_field, signature_field_option, signing_default, signing_default_method, project, contract, document, stored_file
FR-SIGN-002	Coordinate Field Placement	signature_request, signature_request_file, signer, signature_field, signature_field_option, stored_file
FR-SIGN-003	Send Validation Gate	signature_request, signature_request_file, signer, reviewer, signature_field, stored_file, signing_default, company_seal
FR-SIGN-004	Token-Gated Public Signing	signing_token, signer, reviewer, signature_request, signature_request_event, stored_file, outbound_email
FR-SIGN-005	Responsive Signing Pad	saved_signature, users, stored_file, signer, signature_field
FR-SIGN-006	Company Seal Role Gate	company_seal, company_seal_allowed_role, company_seal_use, role, signature_field, signature_request, signer, stored_file
FR-SIGN-007	Sequential Signing Handoff	signature_request, signer, reviewer, signing_token, signature_request_event, outbound_email
FR-SIGN-008	Recipient Decline Workflow	signature_request, signer, reviewer, signing_token, signature_request_event, outbound_email
FR-SIGN-009	Saved Signature Profile	saved_signature, users, stored_file, signer, signature_field
FR-SIGN-010	PDF Flattening (`pdf-lib`)	signature_request, signature_request_file, signer, signature_field, stored_file, signature_evidence
FR-SIGN-011	Background Sweeper Daemon	signature_request, signer, reviewer, signing_token, outbound_email, signature_request_event, stored_file
FR-SIGN-012	Immutable Agreement Audit	signature_request, signature_request_event, signature_evidence, stored_file, signer, reviewer, activity_log
FR-SIGN-013	Advanced Electronic Signature Assurance	signature_request, signer, signing_token, signature_evidence, signature_request_event, stored_file
FR-SIGN-014	Reviewer Approval Before Finalisation	signature_request, reviewer, signature_request_event, outbound_email
FR-TEAM-001	Team Roster Management	team, team_member, org_member, department, team_project, project
FR-TEAM-002	Member Capacity Hours	org_member, work_session, task, time_off_request
FR-TEAM-003	Time-Off Request Lifecycle	time_off_request, org_member, notification, activity_log
FR-TEAM-004	Meeting Scheduling	meeting, meeting_participant, meeting_agenda_item, calendar_event, calendar_event_attendee, notification
FR-TEAM-005	Company Announcements	announcement, announcement_department, announcement_read, department, notification
FR-TEAM-006	Team Activity Stream	activity_log, notification, task_status_change,

		work_session, meeting, announcement
FR-TEAM-007	Member Directory Search	users, org_member, department, team, team_member, org_member_skill
FR-TEAM-008	Team Access Levels and Member Management	org_member, role, invitation, invitation_project, project_member, department, team_member, activity_log
FR-TEAM-009	Custom Role Creation and Editing	role, role_template, org_member, invitation, activity_log
FR-REP-001	Portfolio Health Overview	project, project_member, project_goal, phase, epic, story, task, board, board_column, work_session, baseline_change, activity_log
FR-REP-002	Time Tracking & AI Share	work_session, task, agent_event, agent_token_usage, org_member
FR-REP-003	Schedule Slippage Report	project, phase, epic, story, task, baseline_change
FR-REP-004	Team Workload Heatmap	org_member, work_session, task, time_off_request, team, team_member
FR-REP-005	Agreement Velocity Metrics	signature_request, signer, signature_request_event, contract
FR-REP-006	System Audit Log Viewer	activity_log, signature_request_event, platform_activity_log, platform_admin, users
FR-SETT-001	Organization Branding	organization, stored_file
FR-SETT-002	Member Role Administration	org_member, role, invitation, activity_log
FR-SETT-003	Company Seal Configuration	company_seal, company_seal_allowed_role, company_seal_use, role, stored_file
FR-SETT-004	Global Signing Defaults	signing_default, signing_default_method
FR-SETT-005	Notification Preferences	user_preference, notification_event, user_notification_setting
FR-SETT-006	Demo State Reset Utility	users, session, organization, org_member, role, project, project_member, platform_activity_log, activity_log